

What is DevStack

DevStack คือ ชุดติดตั้ง OpenStack แบบ All-in-One สำหรับการเรียนรู้, Lab และ Development

เปรียบเทียบ :

ชื่อ	จุดประสงค์
DevStack	สำหรับทดลอง, เรียนรู้ และ PoC
OpenStack	สำหรับใช้งานจริงในองค์กร

DevStack จะติดตั้ง Service หลักของ OpenStack ให้ครบในเครื่องเดียว เช่น

- Keystone (Identity)
- Nova (Compute)
- Neutron (Network)
- Glance (Image)
- Cinder (Block Storage)
- Horizon (Web UI)
- Placement
- RabbitMQ
- MariaDB

How to install “Devstack” for Testing

ขนาดของ VM ที่แนะนำ (สำหรับ Lab เครื่องเดียว)

- vCPU: 4 Core (ขั้นต่ำ)
- RAM: 16 GB (แนะนำ 24-32 GB)
- Disk: 100 GB+
- เปิด Nested Virtualization
- Ubuntu Server 24.04 LTS Minimal Install

1. Update ระบบ

```
sudo apt update  
  
sudo apt upgrade -y  
  
sudo reboot
```

2. ติดตั้ง package พื้นฐาน

```
sudo apt install -y \  
  
git vim curl wget net-tools \  
  
python3-dev python3-pip \  
  
build-essential sudo
```

3. สร้าง User stack และกำหนดสิทธิ์ sudoer

```
sudo useradd -s /bin/bash -d /opt/stack -m stack  
  
echo "stack ALL=(ALL) NOPASSWD: ALL" | \  
  
sudo tee /etc/sudoers.d/stack
```

4. เปลี่ยน Permission ของ path

```
sudo chmod +x /opt/stack
```

Login เป็น stack

```
sudo su - stack
```

5. Clone DevStack

แนะนำ branch 2025.2 [At September 2026]

```
git clone https://opendev.org/openstack/devstack
```

```
cd devstack
```

```
git checkout stable/2025.2
```

6. สร้าง local.conf

```
cat > local.conf << EOF
```

```
[[local|localrc]]
```

```
ADMIN_PASSWORD=admin
```

```
DATABASE_PASSWORD=admin
```

```
RABBIT_PASSWORD=admin
```

```
SERVICE_PASSWORD=admin
```

```
HOST_IP=<IP_VM>
```

```
enable_service horizon
```

```
EOF
```

ตัวอย่าง *HOST_IP=10.200.124.188*

7. ตรวจสอบ Virtualization

```
egrep -c '(vmx|svm)' /proc/cpuinfo
```

ต้องมากกว่า 0

8. ตรวจสอบ KVM

```
lsmod | grep kvm
```

ควรเห็น `kvm_intel` หรือ `kvm_amd`

9. เริ่มติดตั้ง

```
./stack.sh
```

ปกติใช้เวลา 15 - 40 นาที ขึ้นกับ Internet และ Spec VM

8. หลังติดตั้งเสร็จ

จะเห็นประมาณนี้

Horizon is now available at

<http://10.200.124.188/dashboard>

admin

admin

9. โหลด Environment

```
source openrc
```

10. ทดสอบ list service

```
openstack service list
```

ควรเห็น

- Keystone
- Glance
- Nova
- Neutron
- Placement
- Cinder
- Horizon

11. Allow Large image

```
sudo vi /etc/apache2/mods-available/reqtimeout.conf
```

เพิ่มตัวแปรในไฟล์ ถ้ามีค่าเดิมอยู่ให้ comment ออกด้วย #

```
RequestReadTimeout header=0
```

```
RequestReadTimeout body=0
```

Basic Admin Command

🔗 Service

openstack service list

🔗 Compute Host

openstack hypervisor list

🔗 Network

openstack network list

🔗 Router

openstack router list

🔗 Image

openstack image list

🔗 VM

openstack server list

🔗 Volume

openstack volume list

OpenStack Admin

OpenStack service component

1. Keystone (Identity)
2. Glance (Image)
3. Nova (Compute)
4. Neutron (Network)
5. Cinder (Block Storage)
6. Horizon (Dashboard)
7. Floating IP
8. Security Group
9. Volume Attach
10. Snapshot / Backup

How to create new Network

1. สร้าง Network บน OS ที่ติดตั้ง OpenStack

ค้นหาว่าในเครื่อง OpenStack มีการประกาศ configuration section [ml2_type_flat] และ flat_networks อยู่ไฟล์ไหน

```
sudo grep -R "[ml2_type_flat]" -n /etc/neutron /opt/stack/neutron 2>/dev/null

sudo grep -R "flat_networks" -n /etc/neutron /opt/stack/neutron 2>/dev/null
```

แก้ไขไฟล์ Config ที่หาเจอ

```
sudo vi /etc/neutron/plugins/ml2/ml2_conf.ini

#change from

[ml2_type_flat]

flat_networks = public,

#to

[ml2_type_flat]

flat_networks = public,public124
```

ตรวจสอบเนื้อหา

```
sudo grep -n -A3 "[ml2_type_flat]" /etc/neutron/plugins/ml2/ml2_conf.ini
```

Restart Neutron Service

```
sudo systemctl restart devstack@neutron-api.service
```

ตรวจสอบ Network ทั้งหมดบน OpenStack

```
openstack network list
```

2. สร้าง Network ใหม่บน OpenStack

สร้าง Network:

```
openstack network create \  
  
--external \  
  
--share \  
  
--provider-network-type flat \  
  
--provider-physical-network public124 \  
  
external124
```

3. สร้าง Subnet ใหม่บน OpenStack

สร้าง Subnet:

```
openstack subnet create \  
  
--network external124 \  
  
--subnet-range 10.200.124.0/23 \  
  
--gateway 10.200.124.1 \  
  
--allocation-pool start=10.200.124.190,end=10.200.124.210 \  
  
--dns-nameserver 10.200.124.173 \  
  
external-subnet124
```

4. สร้าง Router ใหม่บน OpenStack

สร้าง router:

```
openstack router create router-external124
```

แสดงรายละเอียด router:

```
openstack router show router-external124
```

5. สร้าง External Gateway ใหม่บน OpenStack โดยใช้ Router

```
openstack router set --external-gateway external124 router-external124
```

แสดงรายละเอียด Router:

```
openstack router show router-external124
```

6. เชื่อมต่อ Private Subnet กับ Router

แสดง Subnet ทั้งหมดเพื่อนำ ID มาใช้ผูกกับ Router

```
openstack subnet list
```

ผูก Subnet กับ Router:

```
openstack router add subnet \  
  
router-external124 \  
  
ca9a1f6d-59d7-4058-8a50-8f2653c2ef2c
```

7. สร้าง floating IP address ใหม่บน OpenStack และ ผูกกับ VM

สร้าง Floating IP:

```
openstack floating ip create external124
```

ผูก Floating IP กับ VM:

```
openstack server add floating ip test-vm124-ssh 10.200.124.198
```

How to create new instances

1. สร้าง instance ใหม่โดยใช้ ISO File

1.1 Upload Image

Upload image โดย Command Line

```
openstack image create "RHEL95" \  
--file rhel-9.5-x86_64-dvd.iso \  
--disk-format iso \  
--container-format bare
```

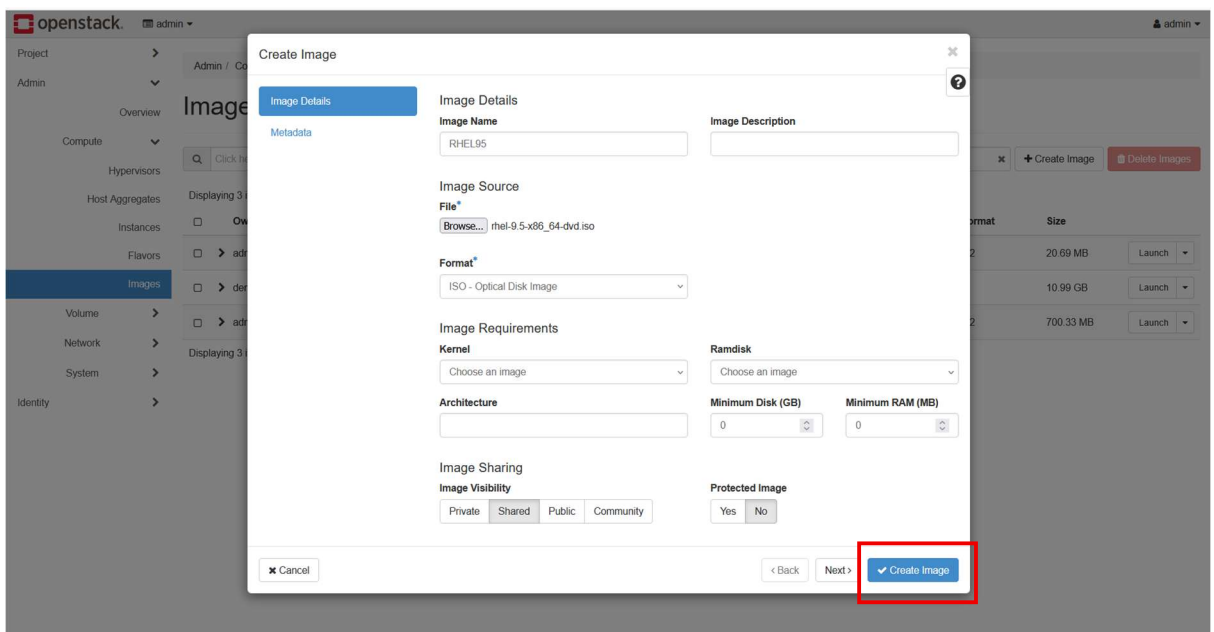
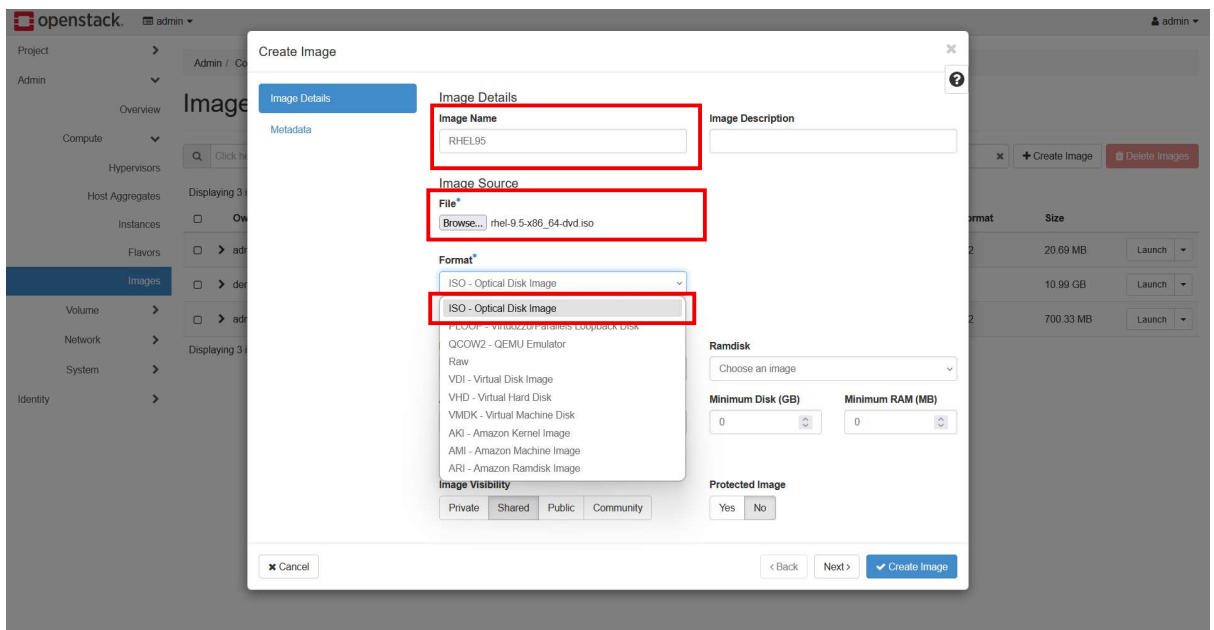
ตรวจสอบ Image

```
openstack image list
```

Upload Image โดยหน้า Web Console

The screenshot shows the OpenStack Web Console interface. The left sidebar contains a navigation menu with options: Project, Admin, Overview, Compute, Hypervisors, Host Aggregates, Instances, Flavors, Images (highlighted), Volume, Network, System, and Identity. The main content area is titled 'Images' and includes a search bar with the text 'Click here for filters or full text search.' To the right of the search bar are two buttons: '+ Create Image' (highlighted with a red box) and 'Delete Images'. Below these buttons, a table displays a list of images. The table has columns for Owner, Name, Type, Status, Visibility, Protected, Disk Format, and Size. Three images are listed: 'cirros-0.6.3-x86_64-disk' (admin), 'redhat9.5' (demo), and 'ubuntu-jammy' (admin). Each row has a 'Launch' button and a dropdown arrow.

Owner	Name	Type	Status	Visibility	Protected	Disk Format	Size
admin	cirros-0.6.3-x86_64-disk	Image	Active	Public	No	QCOW2	20.69 MB
demo	redhat9.5	Image	Active	Image from Other Project - Non-Public	No	ISO	10.99 GB
admin	ubuntu-jammy	Image	Active	Public	No	QCOW2	700.33 MB



1.2 Create Instance

openstack

demo

admin

Project

API Access

Compute

Overview

Instances

Images

Key Pairs

Server Groups

Volumes

Network

Admin

Identity

Project / Compute / Images

Images

Click here for filters or full text search.

Create Image

Delete Images

Displaying 3 items

	Owner	Name	Type	Status	Visibility	Protected	Disk Format	Size	
☐	admin	cirros-0.6.3-x86_64-disk	Image	Active	Public	No	QCOW2	20.69 MB	Launch
☐	demo	redhat9.5	Image	Active	Private	No	ISO	10.99 GB	Launch
☐	admin	ubuntu-jammy	Image	Active	Public	No	QCOW2	700.33 MB	Launch

Displaying 3 items

openstack

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Network

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Identity

Project / Compute / Images

Images

Click here for filters or full text search.

Create Image

Delete Images

Displaying 3 items

	Owner	Name	Type	Status	Visibility	Protected	Disk Format	Size	
☐	admin	cirros-0.6.3-x86_64-disk	Image	Active	Public	No	QCOW2	20.69 MB	Launch
☐	demo	redhat9.5	Image	Active	Private	No	ISO	10.99 GB	Launch
☐	admin	ubuntu-jammy	Image	Active	Public	No	QCOW2	700.33 MB	Launch

Displaying 3 items

Launch Instance

Details

Source

Flavor

Networks

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Please provide the initial hostname for the instance, the availability zone where it will be deployed, and the instance count. Increase the Count to create multiple instances with the same settings.

Project Name

demo

Instance Name

rhel-demo01

Description

Availability Zone

nova

Count

1

Total Instances (10 Max)

30%

2 Current Usage

1 Added

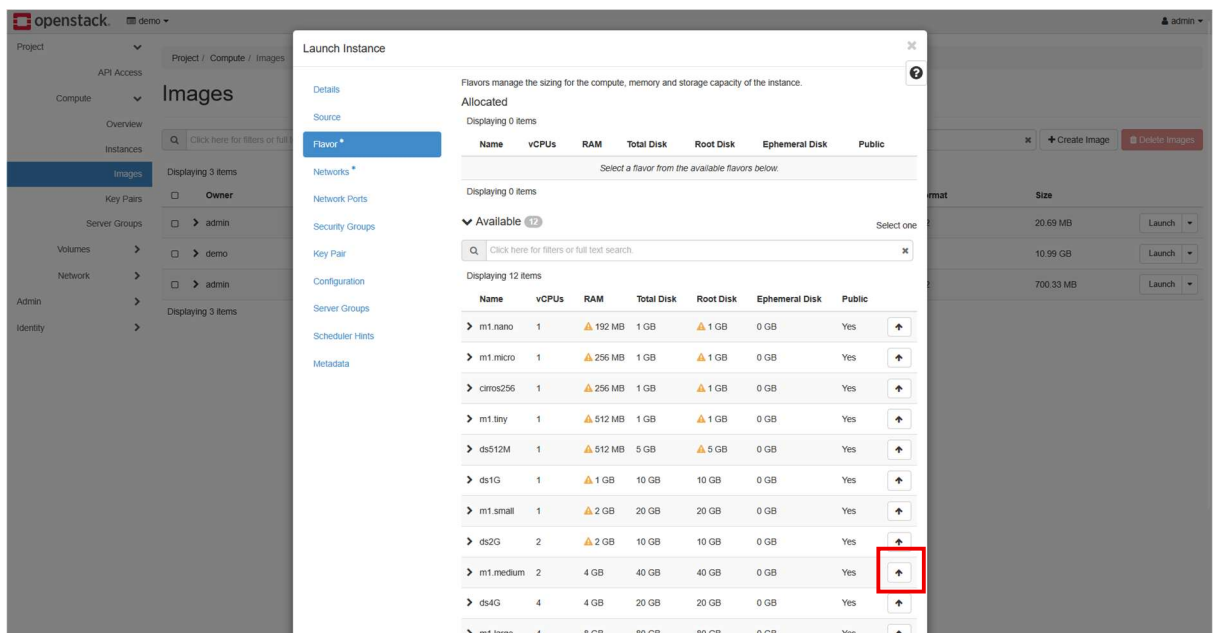
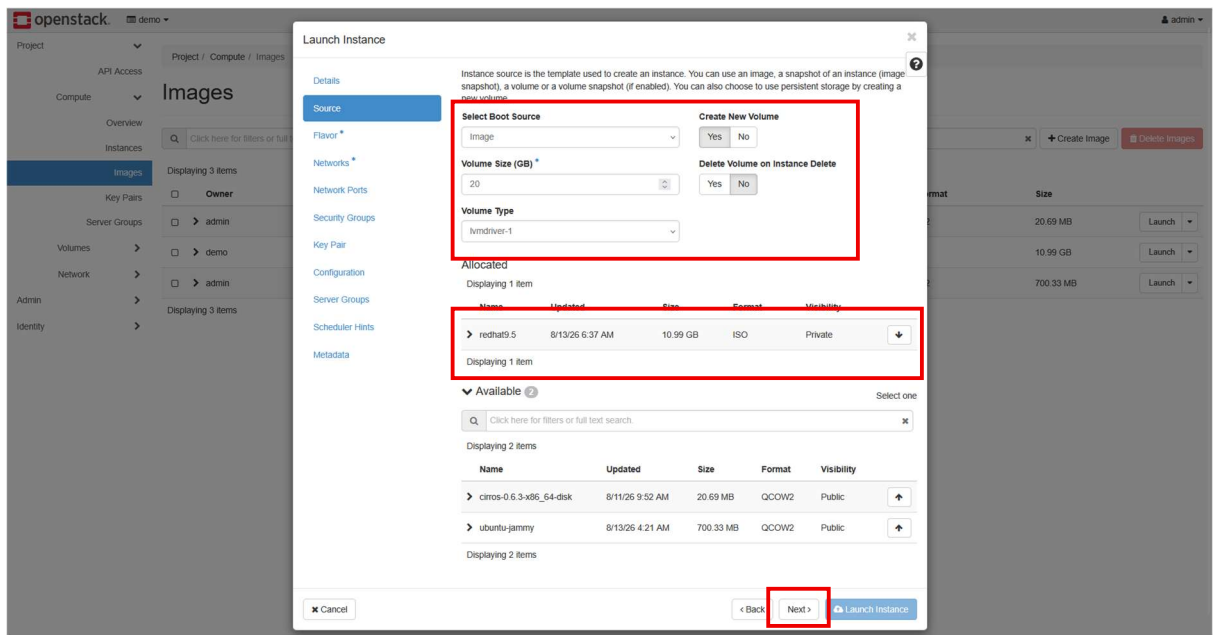
7 Remaining

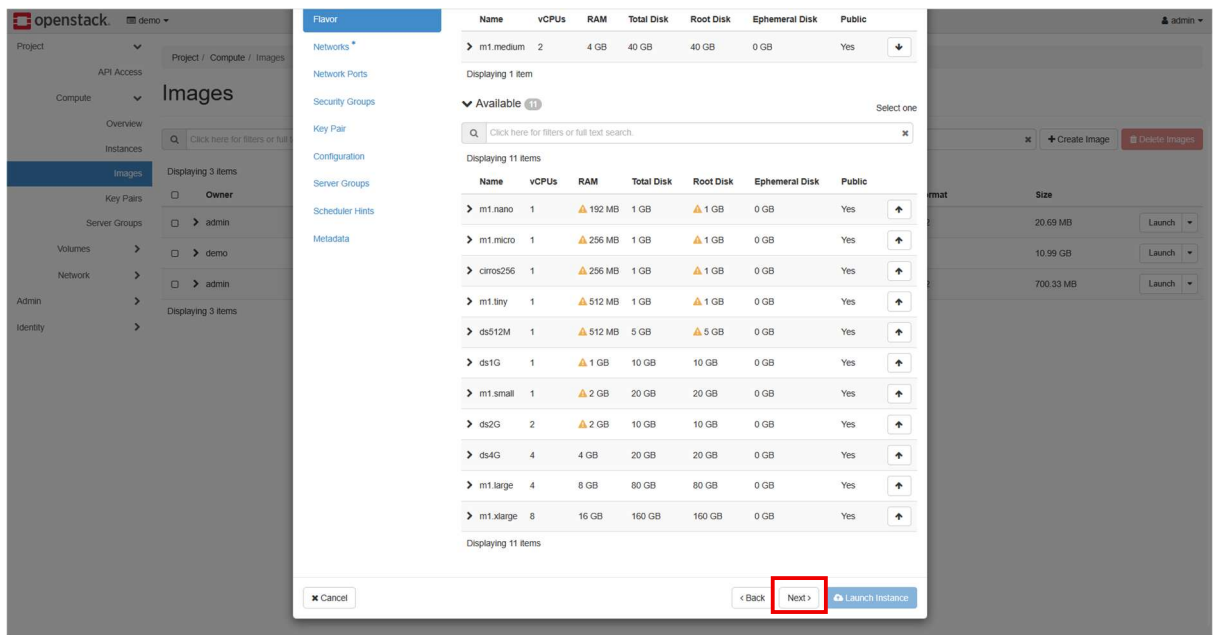
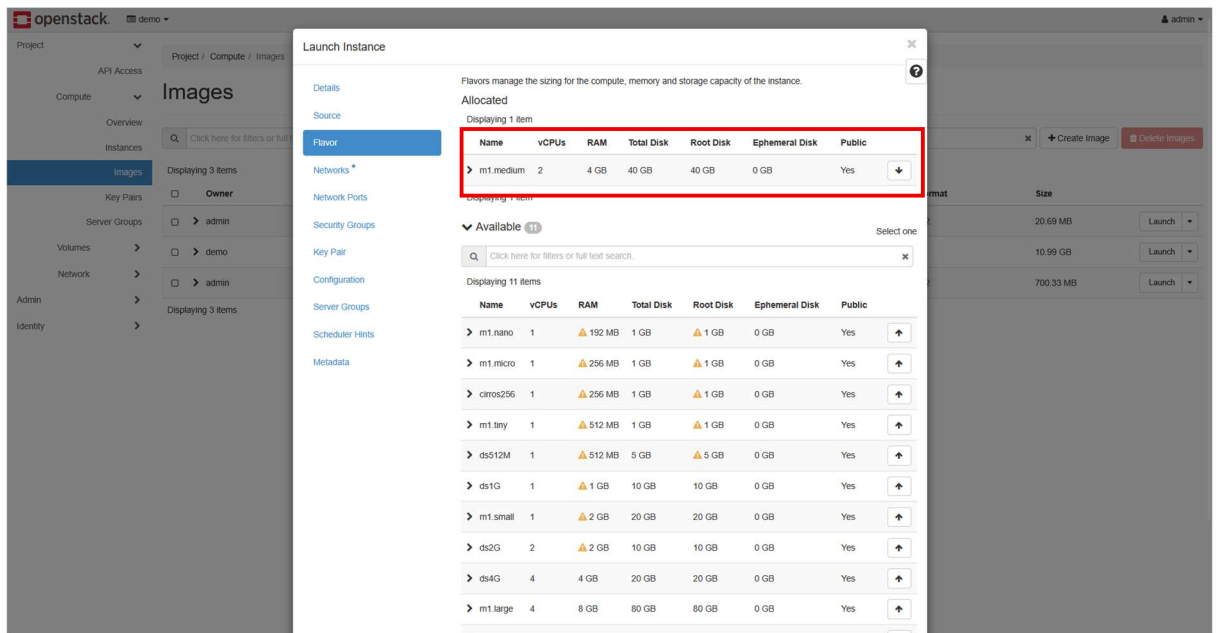
Cancel

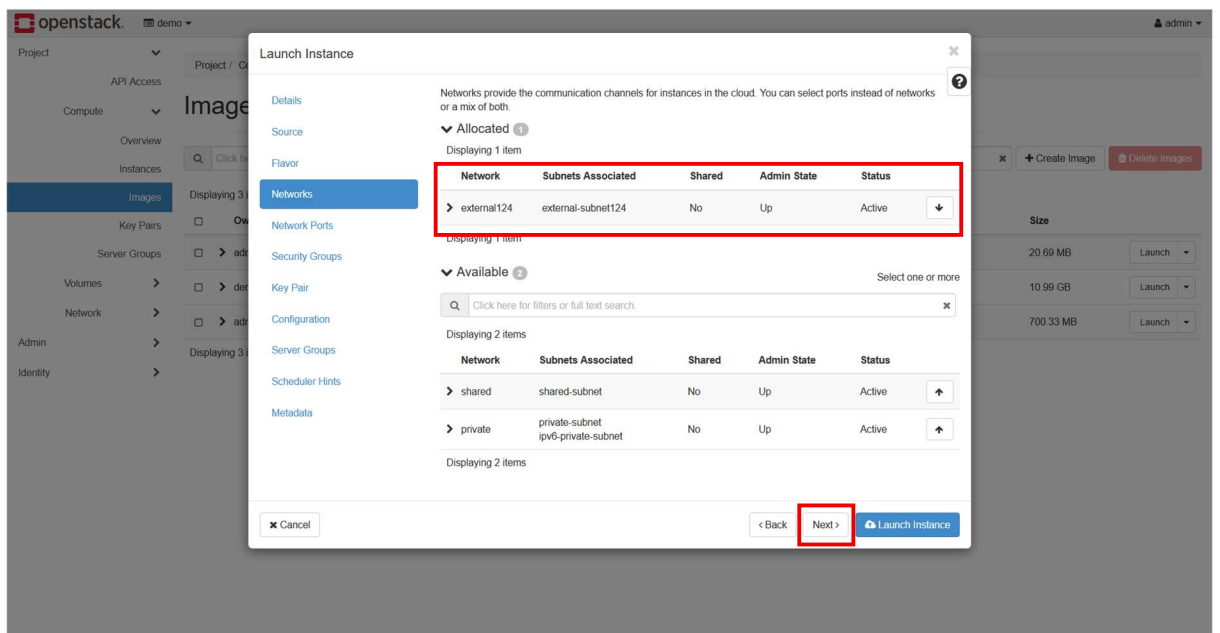
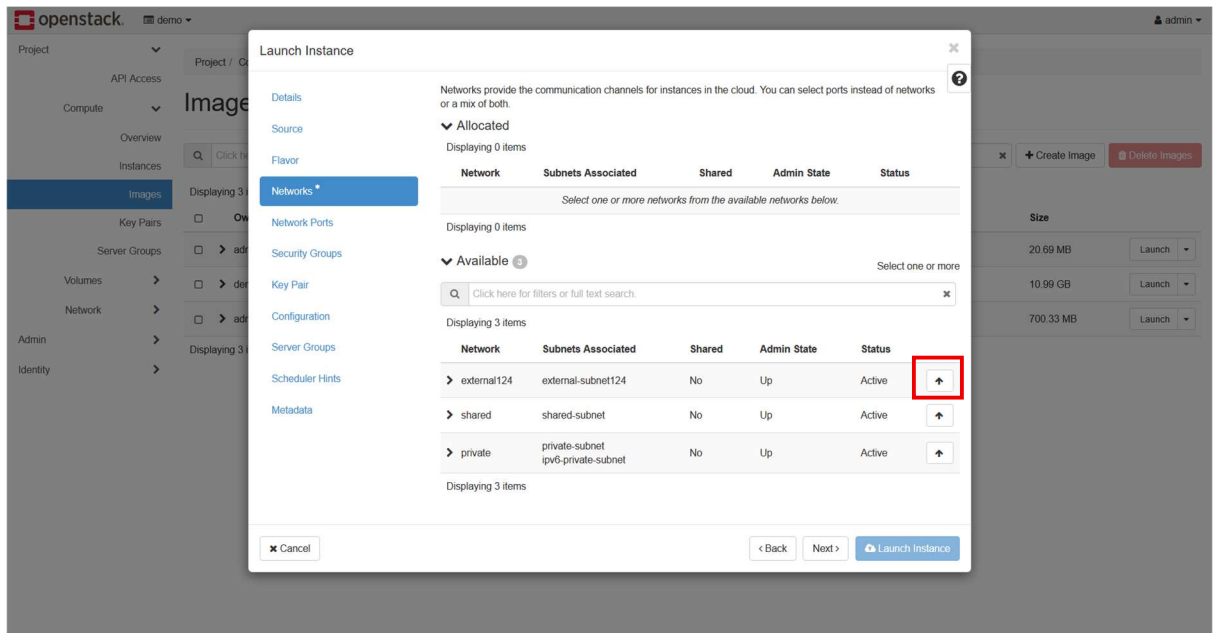
Back

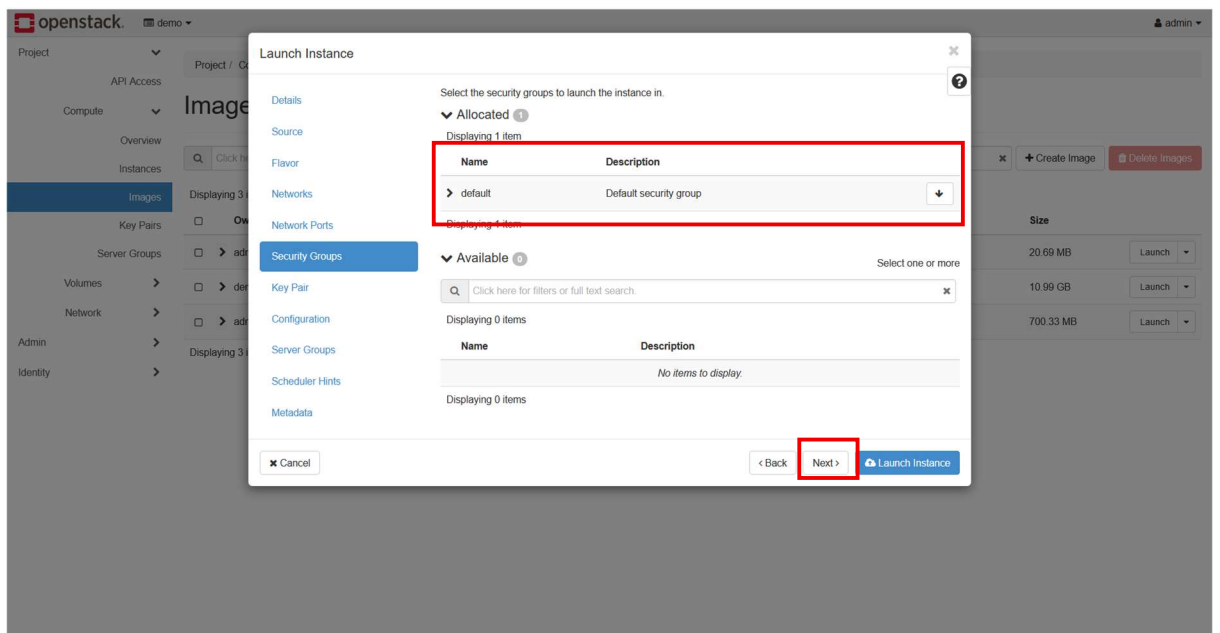
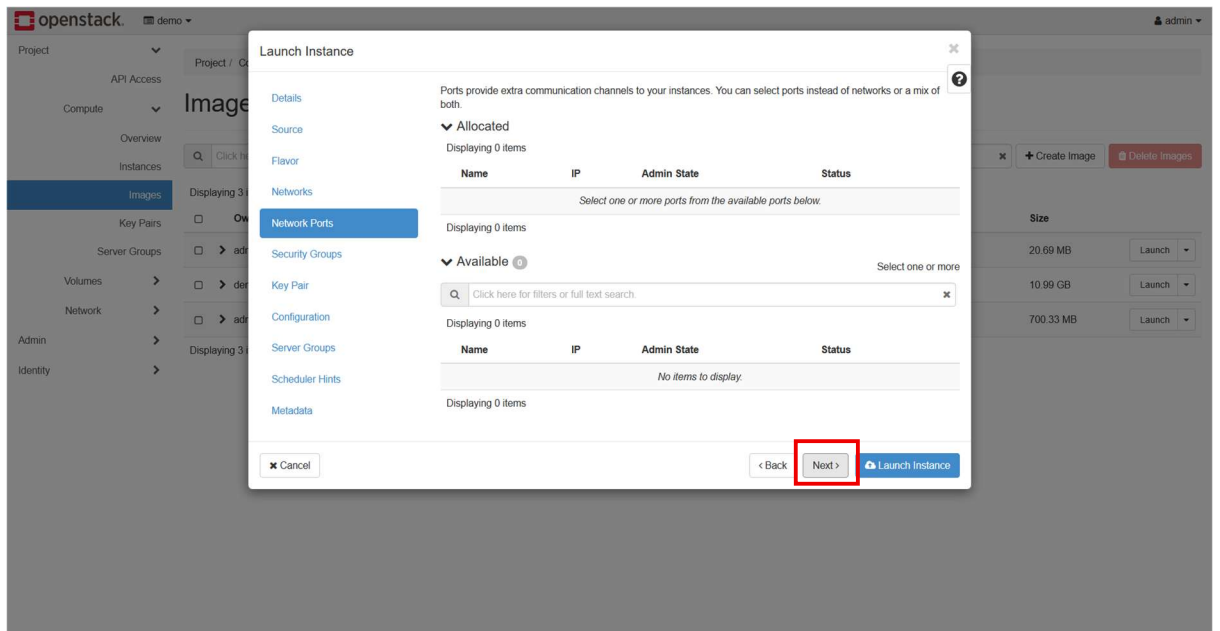
Next

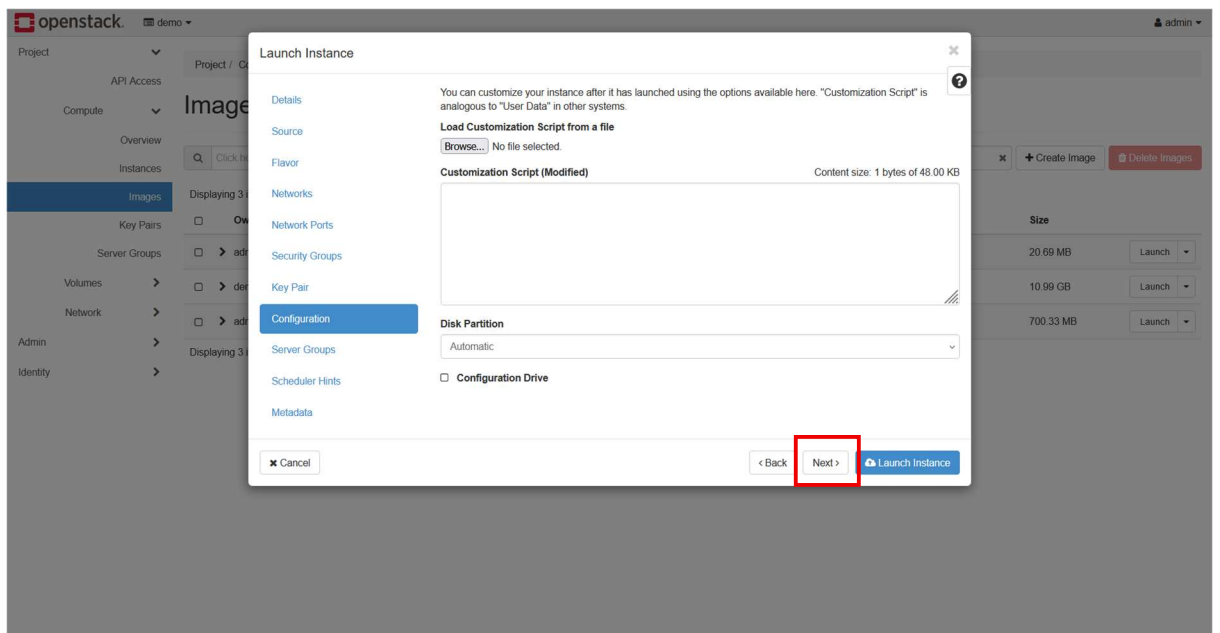
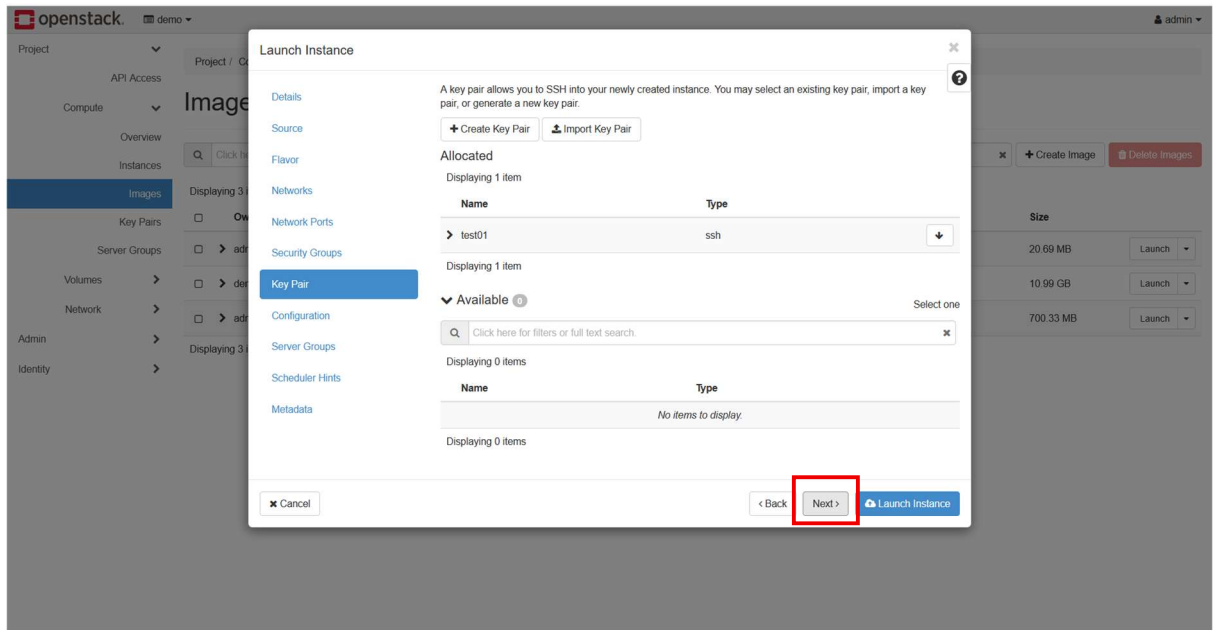
Launch Instance

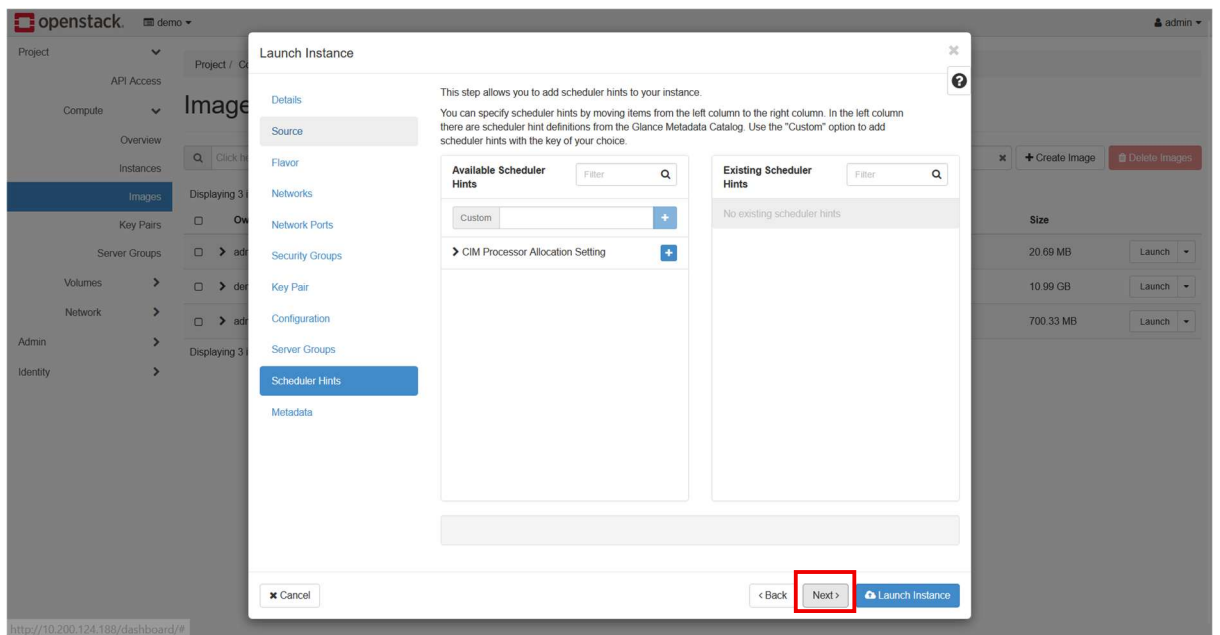
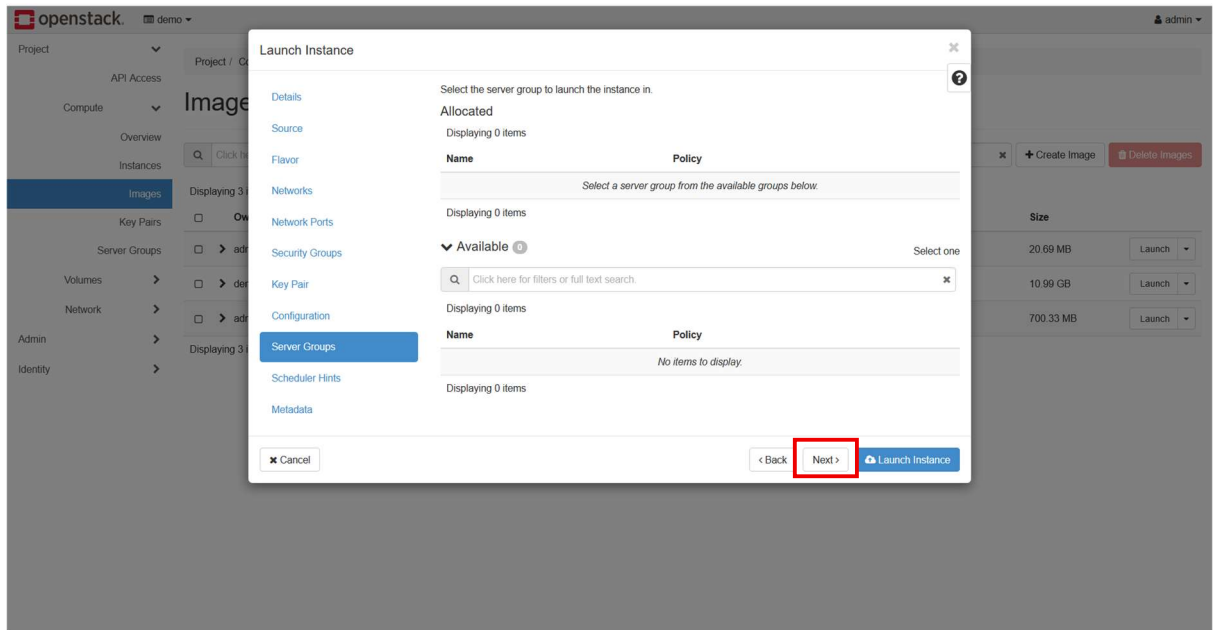


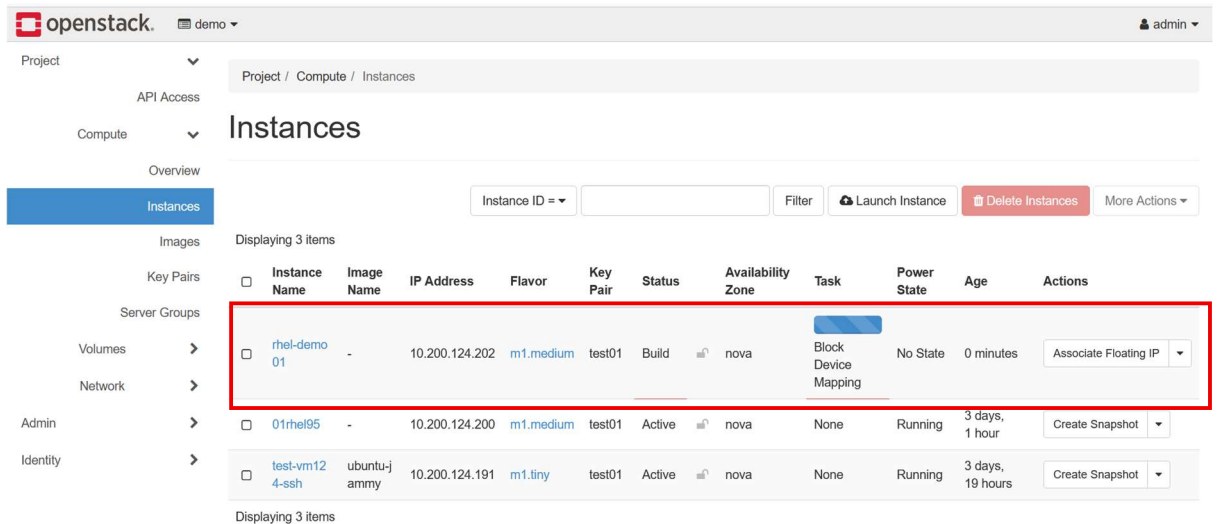
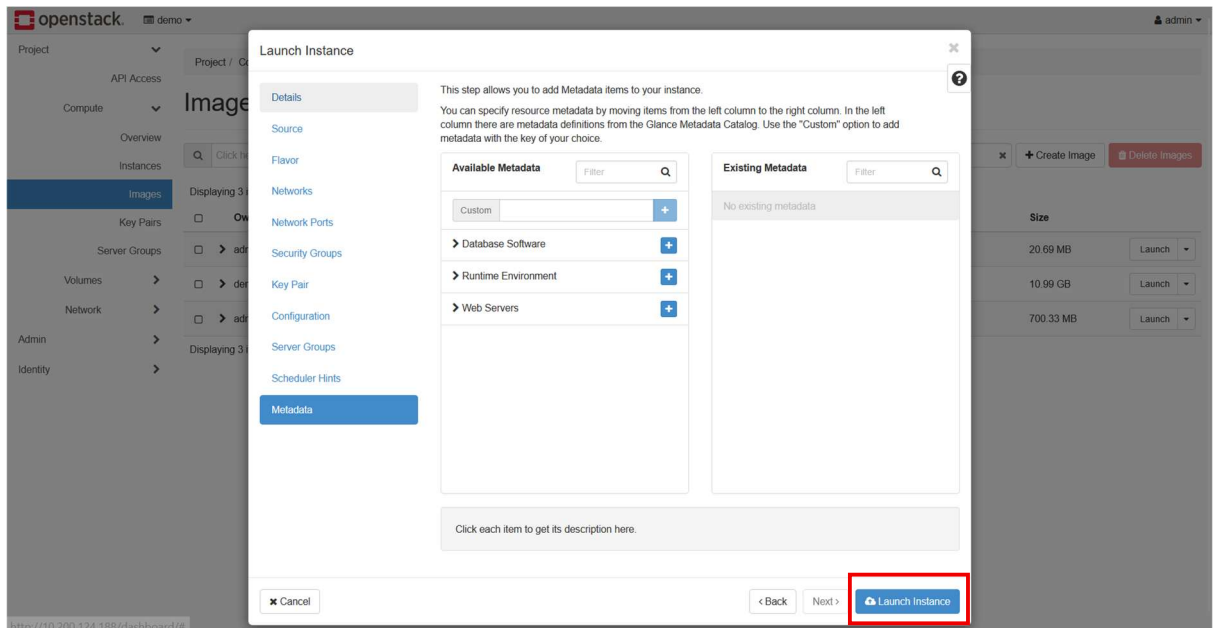












Project

API Access

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Overview

Instances

Images

Key Pairs

Server Groups

Volumes

Network

Admin

Identity

demo

admin

Project / Compute / Instances

Instances

Instance ID = Filter

Launch Instance

Delete Instances

 More Actions

Displaying 3 items

	Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
☐	rhel-demo01	-	10.200.124.202	m1.medium	test01	Active	nova	None	Running	0 minutes	Create Snapshot
☐	01rhel95	-	10.200.124.200	m1.medium	test01	Active	nova	None	Running	3 days, 1 hour	Create Snapshot
☐	test-vm124-sh	ubuntu-jammy	10.200.124.191	m1.tiny	test01	Active	nova	None	Running	3 days, 19 hours	Create Snapshot

Displaying 3 items

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Instances

Instance ID = Filter

Launch Instance

Delete Instances

 More Actions

Displaying 3 items

	Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
☐	rhel-demo01	-	10.200.124.202	m1.medium	test01	Active	nova	None	Running	0 minutes	Create Snapshot
☐	01rhel95	-	10.200.124.200	m1.medium	test01	Active	nova	None	Running	3 days, 1 hour	Create Snapshot
☐	test-vm124-sh	ubuntu-jammy	10.200.124.191	m1.tiny	test01	Active	nova	None	Running	3 days, 19 hours	Create Snapshot

Displaying 3 items

Associate Floating IP

Attach Interface

Detach Interface

Edit Instance

Attach Volume

Detach Volume

Update Metadata

Edit Security Groups

Edit Port Security Groups

Console

View Log

Rescue Instance

Pause Instance

Suspend Instance

Shelve Instance

Resize Instance

Lock Instance

Soft Reboot Instance

Hard Reboot Instance

Migrate Instance

Shut Off Instance

Rebuild Instance

Delete Instance

http://10.200.124.188/dashboard/project/instances/c59b3399-c7db-45ad-9227-225b87011810/?tab=instance_details_console

openstack demo admin

Project / Compute / Instances / rhel-demo01

rhel-demo01 Create Snapshot

Overview Interfaces Log Console Action Log

Instance Console

If console is not responding to keyboard input: click the grey status bar below. To exit the fullscreen mode, click the browser's back button. [Click here to show only console](#)

Connected to QEMU (instance-0000000e) Send Ctrl+Alt+Del

```
[ OK ] Started Show Plymouth Boot Screen.
[ OK ] Started Forward Password Requests to Plymouth Directory Watch.
[ OK ] Reached target Local Encrypted Volumes.
[ OK ] Reached target Path Units.
[ OK ] Finished Wait for udev To Complete Device Initialization.
Starting Device-Mapper Multipath Device Controller...
[ OK ] Started Device-Mapper Multipath Device Controller.
[ OK ] Reached target Preparation for Local File Systems.
[ OK ] Reached target Local File Systems.
[ OK ] Reached target System Initialization.
[ OK ] Reached target Basic System.
/dev/sr0: 8b5a2eb36801d5989447a39bc3810fc9
Fragment sums: b943c9da15f9b87fbf24d26ec7cb886efcd4659ce8dffd8a41cec1f229f8
Fragment count: 20
Supported ISO: yes
Press [Esc] to abort check.
Checking: 019.8%
```

Connected to QEMU (instance-0000000e) Send Ctrl+Alt+Del

Red Hat

RED HAT ENTERPRISE LINUX 9.5 INSTALLATION

us ช่วยด้วย!

WELCOME TO RED HAT ENTERPRISE LINUX 9.5.

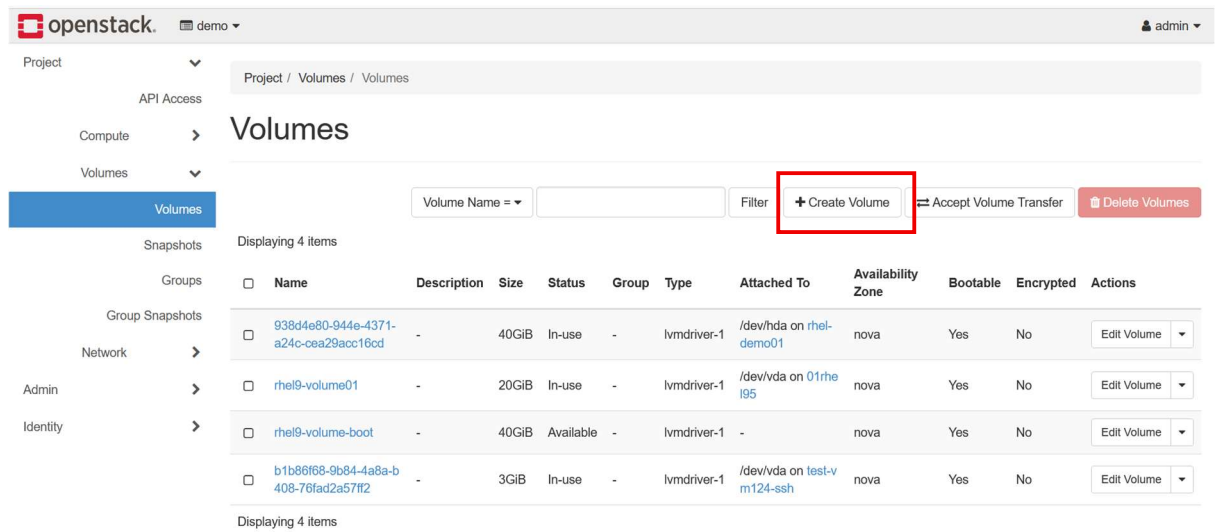
คุณต้องการใช้ภาษาอะไรในการติดตั้ง?

ไทย	Thai	ไทย (ไทย)
العربية	Arabic	
English	English	
Français	French	
Deutsch	German	
日本語	Japanese	
中文	Mandarin Chinese	
Русский	Russian	
Español	Spanish	
Afrikaans	Afrikaans	
አማርኛ	Amharic	
অসমীয়া	Assamese	

Quit Continue

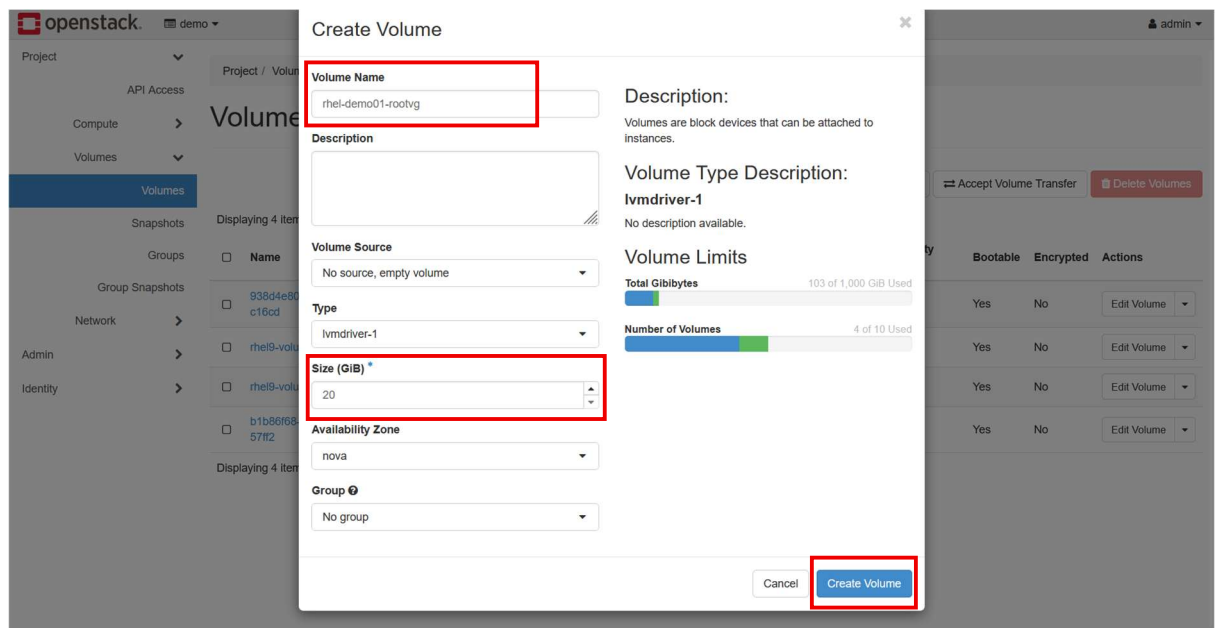
ติดตั้ง OS ตามปกติจนถึงขั้นตอนการ scan disk

กลับมาที่หน้า OpenStack เพื่อสร้าง disk สำหรับติดตั้ง OS



The screenshot shows the OpenStack Volumes dashboard. The left sidebar contains navigation links for Project, API Access, Compute, Volumes, Snapshots, Groups, Group Snapshots, Network, Admin, and Identity. The main content area is titled 'Volumes' and displays a table of existing volumes. A red box highlights the '+ Create Volume' button in the top right corner of the dashboard.

Name	Description	Size	Status	Group	Type	Attached To	Availability Zone	Bootable	Encrypted	Actions
938d4e80-944e-4371-a24c-caa29acc16cd	-	40GiB	In-use	-	lvmdriver-1	/dev/hda on rhel-demo01	nova	Yes	No	Edit Volume
rhel9-volume01	-	20GiB	In-use	-	lvmdriver-1	/dev/vda on 01rhel95	nova	Yes	No	Edit Volume
rhel9-volume-boot	-	40GiB	Available	-	lvmdriver-1	-	nova	Yes	No	Edit Volume
b1b86f68-9b84-4a8a-b408-76fad2a57ff2	-	3GiB	In-use	-	lvmdriver-1	/dev/vda on test-v m124-ssh	nova	Yes	No	Edit Volume



The screenshot shows the 'Create Volume' dialog box. The 'Volume Name' field is set to 'rhel-demo01-rootvg'. The 'Size (GiB)' field is set to '20'. The 'Create Volume' button is highlighted with a red box. The dialog also shows the 'Volume Type' as 'lvmdriver-1' and the 'Availability Zone' as 'nova'.

Volume Name: rhel-demo01-rootvg

Description:

Volume Source: No source, empty volume

Type: lvmdriver-1

Size (GiB): 20

Availability Zone: nova

Group: No group

Volume Limits:

- Total Gibibytes: 103 of 1,000 GiB Used
- Number of Volumes: 4 of 10 Used

Create Volume

openstack

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admin

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Snapshots

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Group Snapshots

Network

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Identity

Project / Volumes / Volumes

Volumes

Volume Name = Filter + Create Volume Accept Volume Transfer Delete Volumes

Displaying 5 items

	Name	Description	Size	Status	Group	Type	Attached To	Availability Zone	Bootable	Encrypted	Actions
☐	rhel-demo01-rootvg	-	20GiB	Available	-	lvmdriver-1	-	nova	No	No	Edit Volume
☐	938d4e80-944e-4371-a24c-cea29ac c16cd	-	40GiB	In-use	-	lvmdriver-1	/dev/hda on rhel-demo01	nova	Yes	No	Edit Volume
☐	rhel9-volume01	-	20GiB	In-use	-	lvmdriver-1	/dev/vda on 01rhel95	nova	Yes	No	Edit Volume
☐	rhel9-volume-boot	-	40GiB	Available	-	lvmdriver-1	-	nova	Yes	No	Edit Volume
☐	b1b86f68-9b84-4a8a-b408-76fad2a 57ff2	-	3GiB	In-use	-	lvmdriver-1	/dev/vda on test-vm124-ssh	nova	Yes	No	Edit Volume

Displaying 5 items

openstack

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Project / Volumes / Volumes

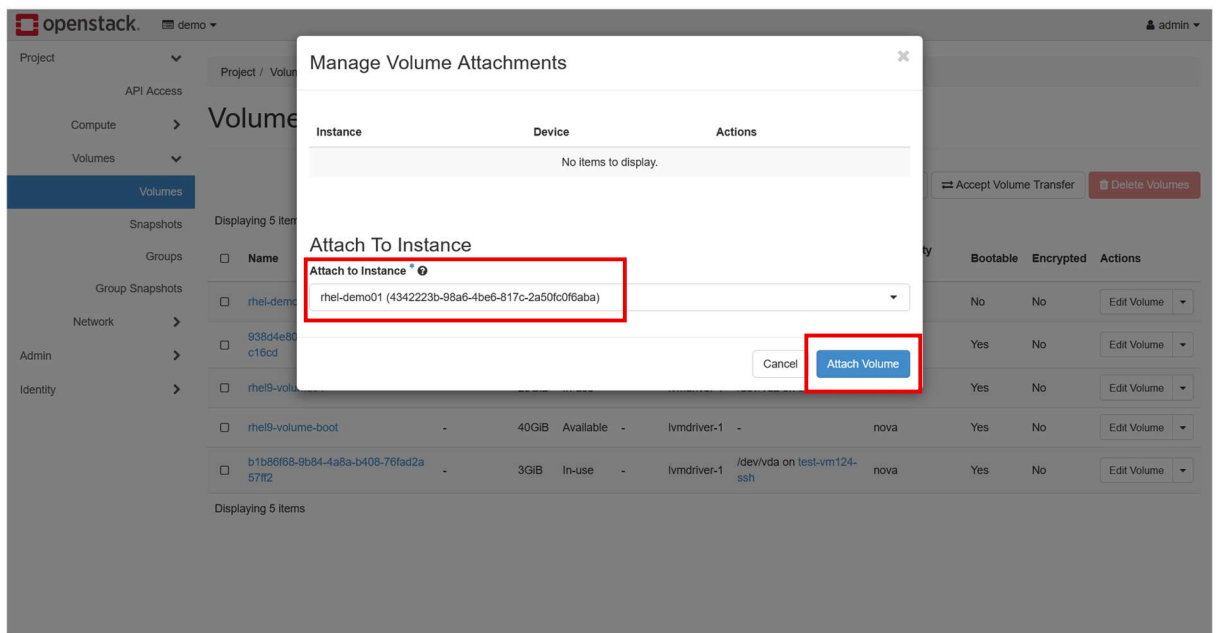
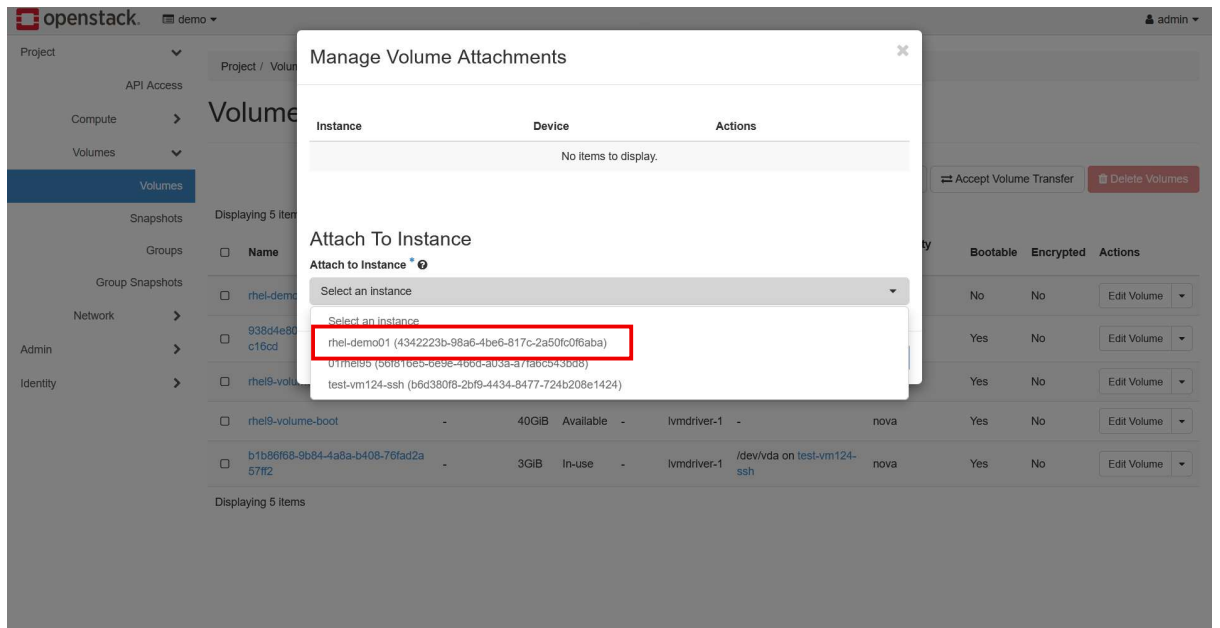
Volumes

Volume Name = Filter + Create Volume Accept Volume Transfer Delete Volumes

Displaying 5 items

	Name	Description	Size	Status	Group	Type	Attached To	Availability Zone	Bootable	Encrypted	Actions
☐	rhel-demo01-rootvg	-	20GiB	Available	-	lvmdriver-1	-	nova	No	No	Edit Volume
☐	938d4e80-944e-4371-a24c-cea29ac c16cd	-	40GiB	In-use	-	lvmdriver-1	/dev/hda on rhel-demo01	nova	Yes	No	Extend Volume Manage Attachments Create Snapshot Change Volume Type Upload to Image Create Volume Transfer Delete Volume Update Metadata
☐	rhel9-volume01	-	20GiB	In-use	-	lvmdriver-1	/dev/vda on 01rhel95	nova	Yes	No	
☐	rhel9-volume-boot	-	40GiB	Available	-	lvmdriver-1	-	nova	Yes	No	
☐	b1b86f68-9b84-4a8a-b408-76fad2a 57ff2	-	3GiB	In-use	-	lvmdriver-1	/dev/vda on test-vm124-ssh	nova	Yes	No	

Displaying 5 items



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Volumes

Volume Name = Filter + Create Volume Accept Volume Transfer Delete Volumes

Displaying 5 items

☐	Name	Description	Size	Status	Group	Type	Attached To	Availability Zone	Bootable	Encrypted	Actions
☐	rhel-demo01-rootvg	-	20GiB	In-use	-	lvmdriver-1	/dev/vda on rhel-demo01	nova	No	No	Edit Volume
☐	938d4e80-944e-4371-a24c-cea29ac16cd	-	40GiB	In-use	-	lvmdriver-1	/dev/hda on rhel-demo01	nova	Yes	No	Edit Volume
☐	rhel9-volume01	-	20GiB	In-use	-	lvmdriver-1	/dev/vda on 01rhel95	nova	Yes	No	Edit Volume
☐	rhel9-volume-boot	-	40GiB	Available	-	lvmdriver-1	-	nova	Yes	No	Edit Volume
☐	b1b86f68-9b84-4a8a-b408-76fad2a57ff2	-	3GiB	In-use	-	lvmdriver-1	/dev/vda on test-vm124-ssh	nova	Yes	No	Edit Volume

Displaying 5 items

Connected to QEMU (instance-00000001)

Send Ctrl+Alt+Del

INSTALLATION DESTINATION

RED HAT ENTERPRISE LINUX 9.5 INSTALLATION

Done us Help!

Device Selection

Select the device(s) you'd like to install to. They will be left untouched until you click on the main menu's "Begin Installation" button.

Local Standard Disks

Specialized & Network Disks

Add a disk...

Storage Configuration

☒ Automatic ☐ Custom

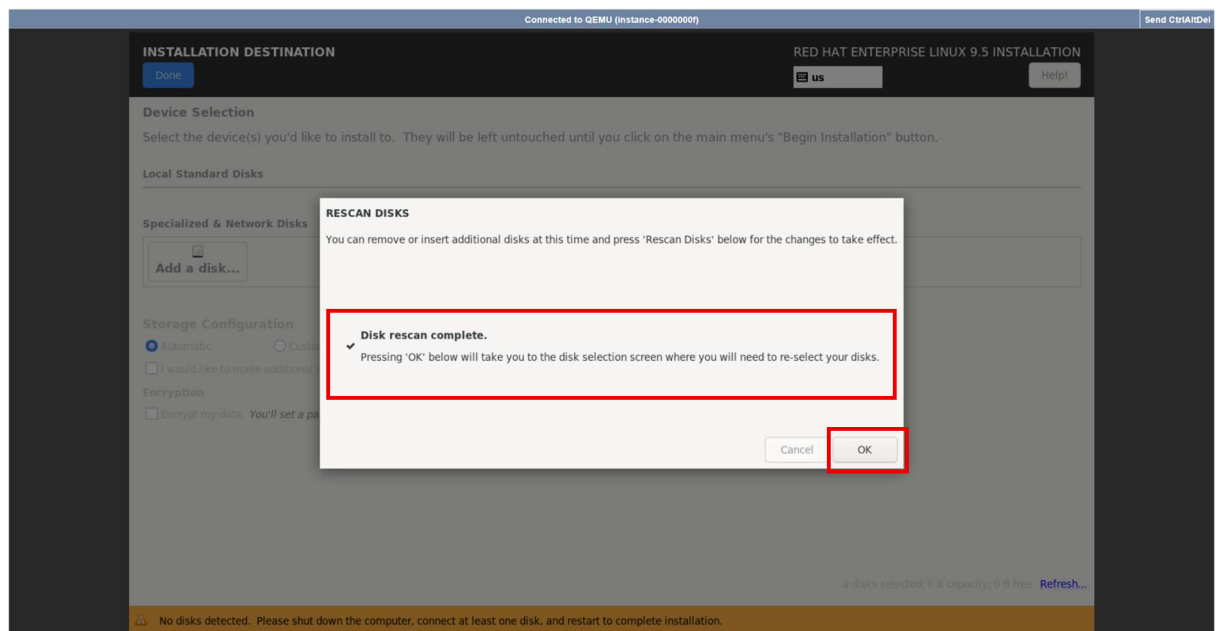
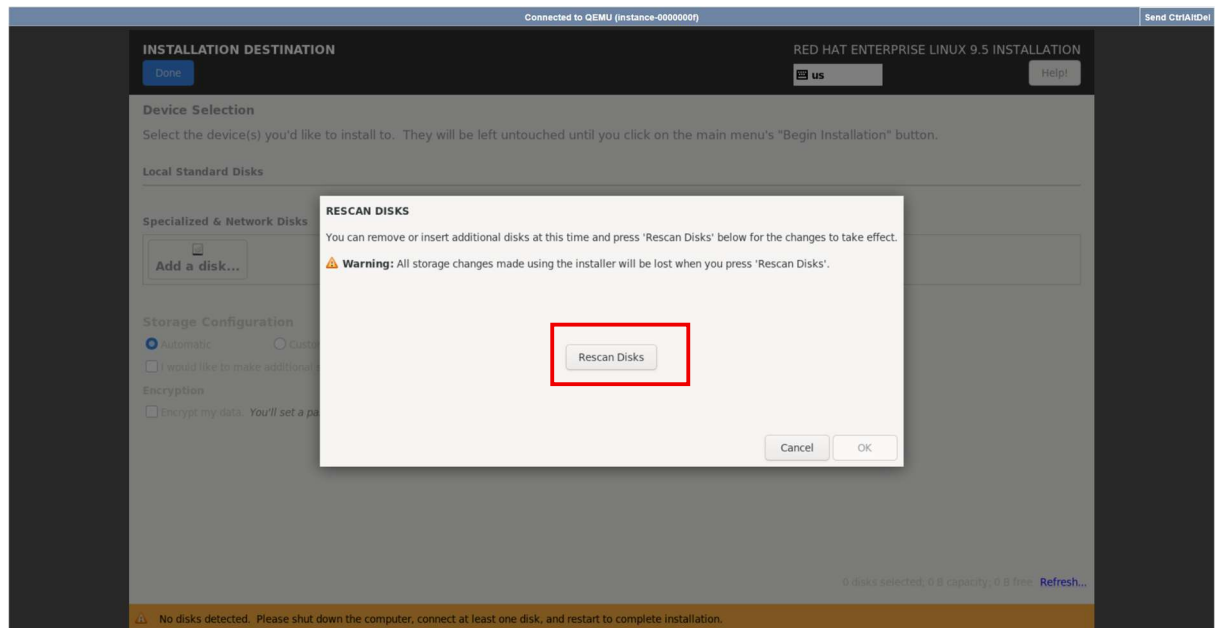
☐ I would like to make additional space available.

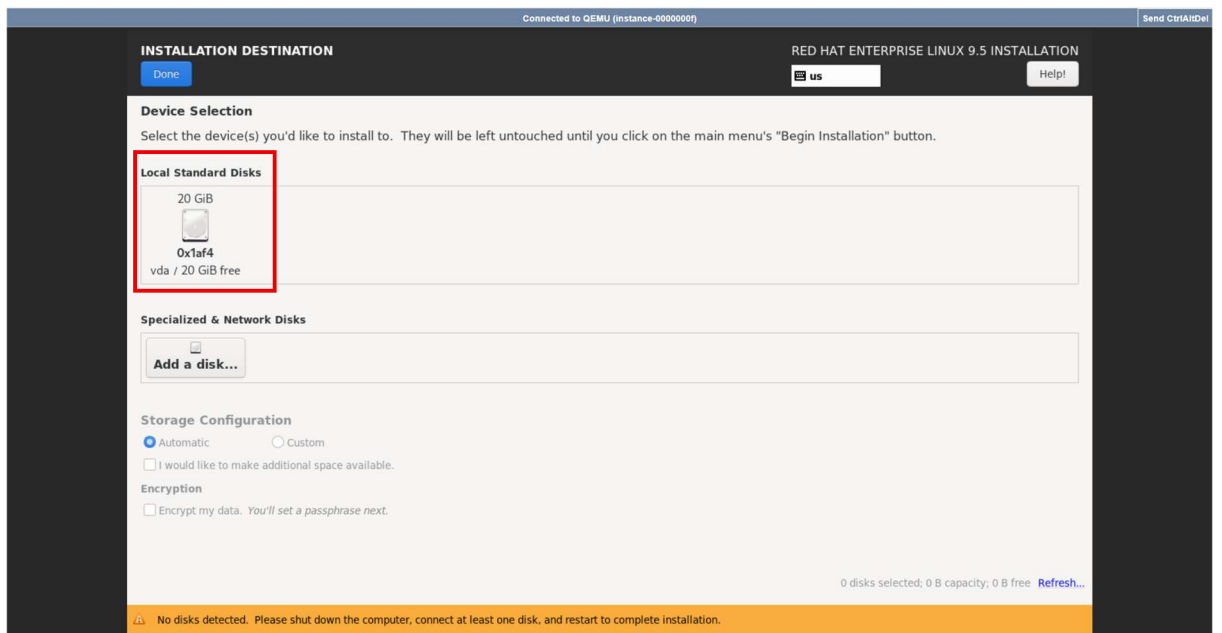
Encryption

☐ Encrypt my data. You'll set a passphrase next.

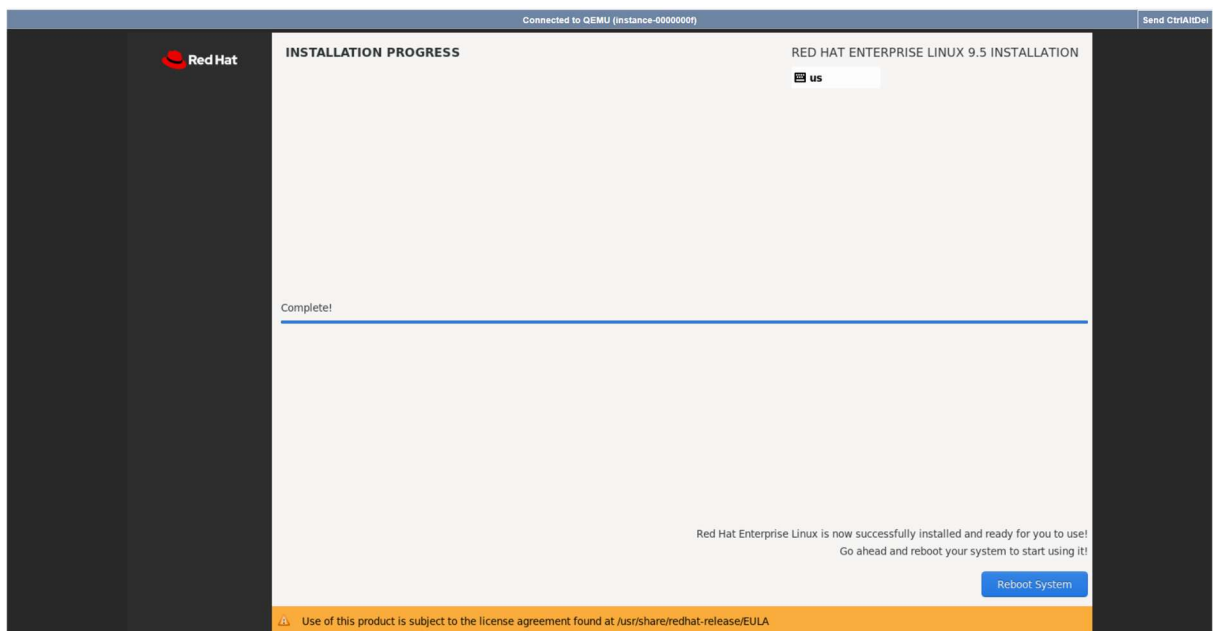
0 disks selected; 0 B capacity; 0 B free Refresh...

No disks detected. Please shut down the computer, connect at least one disk, and restart to complete installation.





ติดตั้ง OS ต่อตามปกติจนเสร็จ



Shutdown OS ที่ติดตั้งเสร็จแล้ว

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Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
rhel-demo01	redhat9.5	10.200.124.193	m1.medium	test01	Active	nova	None	Running	27 minutes	Create Snapshot
01rhel95	-	10.200.124.200	m1.medium	test01	Active	nova	None	Running	3 days, 2 hours	Associate Floating IP Attach Interface Detach Interface Edit Instance Attach Volume Detach Volume Update Metadata Edit Security Groups Edit Port Security Groups Console View Log Rescue Instance Pause Instance Suspend Instance Shelve Instance Resize Instance Lock Instance Soft Reboot Instance Hard Reboot Instance Migrate Instance Shut Off Instance Rebuild Instance Delete Instance
test-vm124-sh	ubuntu-jammy	10.200.124.191	m1.tiny	test01	Active	nova	None	Running	3 days, 20 hours	

Displaying 3 items

http://10.200.124.188/dashboard/project/instances/4342223b-98a6-4be6-817c-2a50fc0f6aba/?tab=instance_details_console

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Confirm Shut Off Instance

You have selected: "rhel-demo01". Please confirm your selection. The instance(s) will be shut off.

CancelShut Off Instance

Displaying 3 items

Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
rhel-demo01	redhat9.5	10.200.124.193	m1.medium	test01	Active	nova	None	Running	27 minutes	Create Snapshot
01rhel95	-	10.200.124.200	m1.medium	test01	Active	nova	None	Running	3 days, 2 hours	Create Snapshot
test-vm124-sh	ubuntu-jammy	10.200.124.191	m1.tiny	test01	Active	nova	None	Running	3 days, 20 hours	Create Snapshot

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	Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
☐	rhel-demo01	-	10.200.124.193	m1.medium	test01	Shutoff	nova	None	Shut Down	29 minutes	Start Instance
☐	01rhel95	-	10.200.124.200	m1.medium	test01	Active	nova	None	Running	3 days, 2 hours	Create Snapshot
☐	test-vm124-sh	ubuntu-jammy	10.200.124.191	m1.tiny	test01	Active	nova	None	Running	3 days, 20 hours	Create Snapshot

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Displaying 3 items

	Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
☐	rhel-demo01	-	10.200.124.193	m1.medium	test01	Shutoff	nova	None	Shut Down	29 minutes	Start Instance
☐	01rhel95	-	10.200.124.200	m1.medium	test01	Active	nova	None	Running	3 days, 2 hours	Create Snapshot
☐	test-vm124-sh	ubuntu-jammy	10.200.124.191	m1.tiny	test01	Active	nova	None	Running	3 days, 20 hours	Create Snapshot

Displaying 3 items

Create Snapshot

Associate Floating IP

Attach Interface

Detach Interface

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Update Metadata

Edit Port Security Groups

Shelve Instance

Resize Instance

Lock Instance

Hard Reboot Instance

Migrate Instance

Rebuild Instance

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Volumes

Volume Name = Filter [+ Create Volume](#) [⇌ Accept Volume Transfer](#) [Delete Volumes](#)

Displaying 5 items

☐	Name	Description	Size	Status	Group	Type	Attached To	Availability Zone	Bootable	Encrypted	Actions
☐	rhel-demo01-rootvg	-	20GiB	Available	-	lvmdriver-1	-	nova	No	No	Edit Volume
☐	938d4e80-944e-4371-a24c-c8a29acc16cd	-	40GiB	Available	-	lvmdriver-1	-	nova	Yes	No	Edit Volume
☐	rhel9-volume01	-	20GiB	In-use	-	lvmdriver-1	/dev/vda on 01rhel95	nova	Yes	No	Edit Volume
☐	rhel9-volume-boot	-	40GiB	Available	-	lvmdriver-1	-	nova	Yes	No	Edit Volume
☐	b1b86f68-9b84-4a8a-b408-76fad2a57ff2	-	3GiB	In-use	-	lvmdriver-1	/dev/vda on test-v m124-ssh	nova	Yes	No	Edit Volume

Displaying 5 items

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Volumes

Volume Name = Filter [+ Create Volume](#) [⇌ Accept Volume Transfer](#) [Delete Volumes](#)

Displaying 5 items

☐	Name	Description	Size	Status	Group	Type	Attached To	Availability Zone	Bootable	Encrypted	Actions
☐	rhel-demo01-rootvg	-	20GiB	Available	-	lvmdriver-1	-	nova	No	No	Edit Volume
☐	938d4e80-944e-4371-a24c-c8a29acc16cd	-	40GiB	Available	-	lvmdriver-1	-	nova	Yes	No	Edit Volume
☐	rhel9-volume01	-	20GiB	In-use	-	lvmdriver-1	/dev/vda on 01rhel95	nova	Yes	No	Edit Volume
☐	rhel9-volume-boot	-	40GiB	Available	-	lvmdriver-1	-	nova	Yes	No	Edit Volume
☐	b1b86f68-9b84-4a8a-b408-76fad2a57ff2	-	3GiB	In-use	-	lvmdriver-1	/dev/vda on test-v m124-ssh	nova	Yes	No	Edit Volume

Displaying 5 items

Edit Volume

Volume Name

rhel-demo01-rootvg

Description

☒ Bootable

Description:

Modify name and description of a volume.
The "Bootable" flag specifies that this volume can be used to launch an instance.

Cancel

Submit

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Displaying 2 items

Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
01rhel95	-	10.200.124.200	m1.medium	test01	Active	nova	None	Running	3 days, 2 hours	Create Snapshot
test-vm124-sh	ubuntu-jammy	10.200.124.191	m1.tiny	test01	Active	nova	None	Running	3 days, 20 hours	Create Snapshot

Displaying 2 items

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Metadata

Please provide the initial hostname for the instance, the availability zone where it will be deployed, and the instance count. Increase the Count to create multiple instances with the same settings.

Project Name

demo

Instance Name

rhel-demo02

Description

Availability Zone

nova

Count

1

Total Instances (10 Max)

30%

2 Current Usage

1 Added

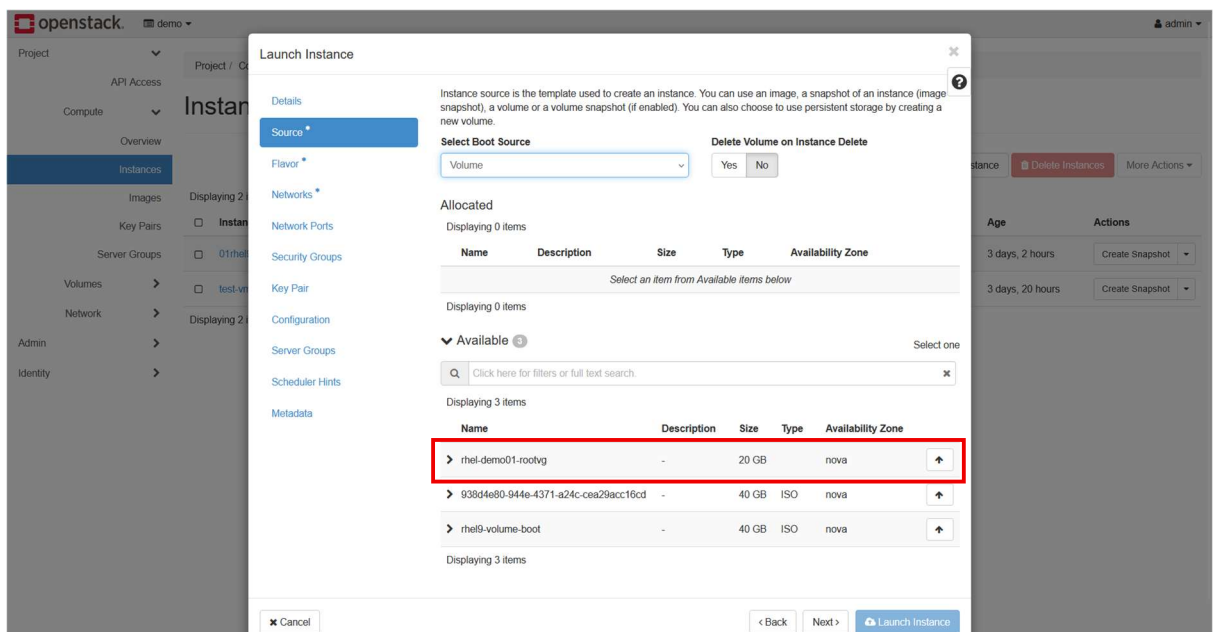
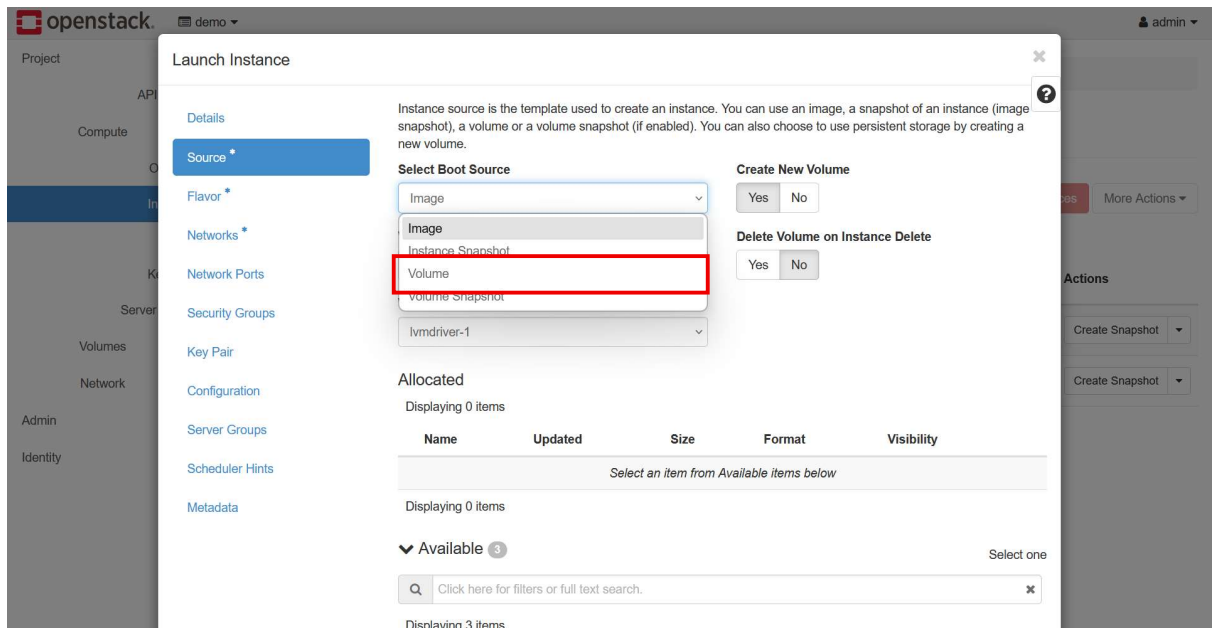
7 Remaining

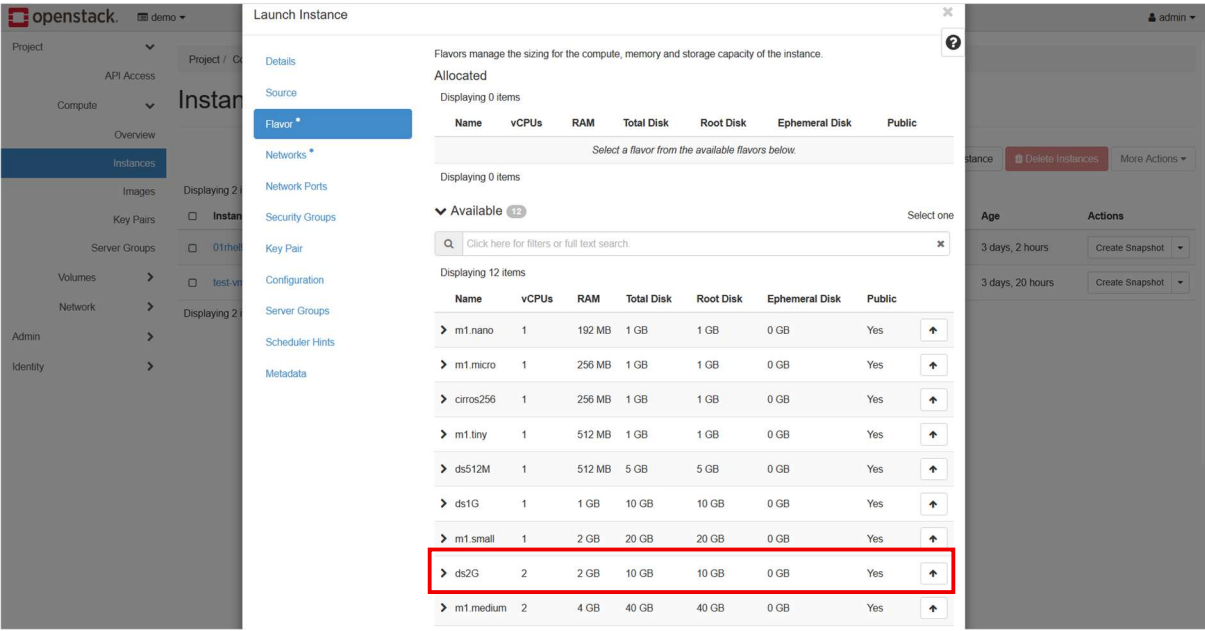
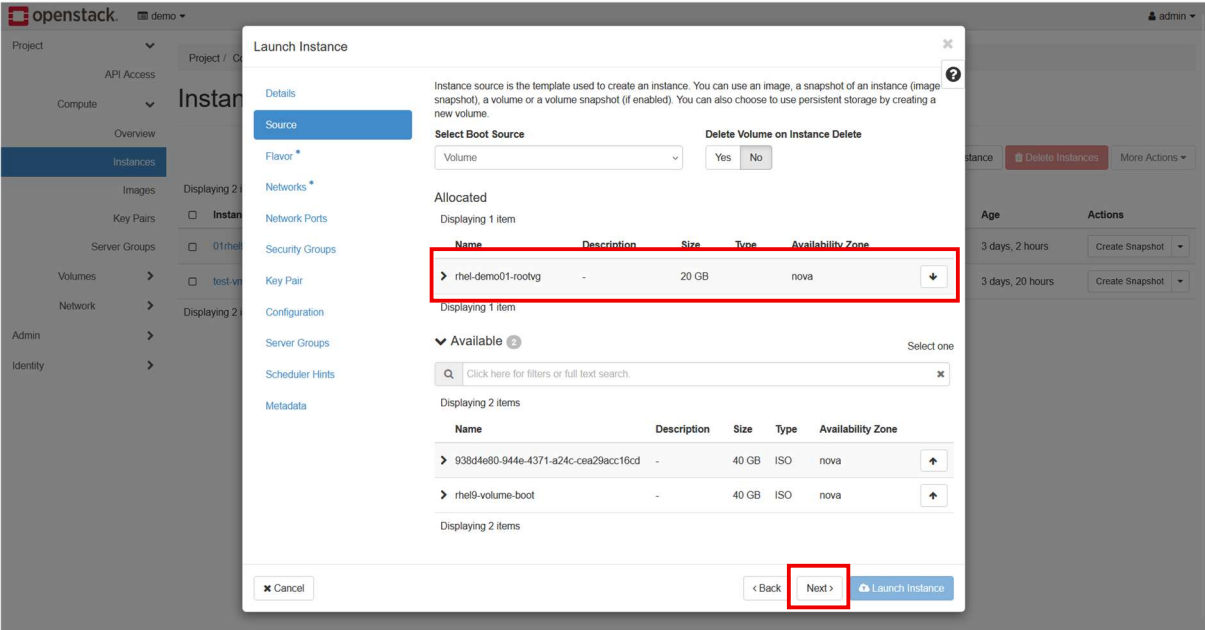
Cancel

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Displaying 2 items

Instance Name

01rhel95

test-vm124-ssh

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Metadata

Flavors manage the sizing for the compute, memory and storage capacity of the instance.

Allocated

Displaying 1 item

Name	vCPUs	RAM	Total Disk	Root Disk	Ephemeral Disk	Public
ds2G	2	2 GB	10 GB	10 GB	0 GB	Yes

Displaying 1 item

Available 11

Click here for filters or full text search

Displaying 11 items

Name	vCPUs	RAM	Total Disk	Root Disk	Ephemeral Disk	Public
m1.nano	1	192 MB	1 GB	1 GB	0 GB	Yes
m1.micro	1	256 MB	1 GB	1 GB	0 GB	Yes
ciros256	1	256 MB	1 GB	1 GB	0 GB	Yes
m1.tiny	1	512 MB	1 GB	1 GB	0 GB	Yes
ds512M	1	512 MB	5 GB	5 GB	0 GB	Yes
ds1G	1	1 GB	10 GB	10 GB	0 GB	Yes
m1.small	1	2 GB	20 GB	20 GB	0 GB	Yes
m1.medium	2	4 GB	40 GB	40 GB	0 GB	Yes
ds4G	4	4 GB	20 GB	20 GB	0 GB	Yes

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Age

3 days, 2 hours

Create Snapshot

3 days, 20 hours

Create Snapshot

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Displaying 2 items

Instance Name

01rhel95

test-vm124-ssh

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Metadata

Flavors manage the sizing for the compute, memory and storage capacity of the instance.

Allocated

Displaying 1 item

Name	vCPUs	RAM	Total Disk	Root Disk	Ephemeral Disk	Public
ds2G	2	2 GB	10 GB	10 GB	0 GB	Yes

Displaying 1 item

Available 11

Click here for filters or full text search

Displaying 11 items

Name	vCPUs	RAM	Total Disk	Root Disk	Ephemeral Disk	Public
m1.nano	1	192 MB	1 GB	1 GB	0 GB	Yes
m1.micro	1	256 MB	1 GB	1 GB	0 GB	Yes
ciros256	1	256 MB	1 GB	1 GB	0 GB	Yes
m1.tiny	1	512 MB	1 GB	1 GB	0 GB	Yes
ds512M	1	512 MB	5 GB	5 GB	0 GB	Yes
ds1G	1	1 GB	10 GB	10 GB	0 GB	Yes
m1.small	1	2 GB	20 GB	20 GB	0 GB	Yes
m1.medium	2	4 GB	40 GB	40 GB	0 GB	Yes
ds4G	4	4 GB	20 GB	20 GB	0 GB	Yes
m1.large	4	8 GB	80 GB	80 GB	0 GB	Yes
m1.xlarge	8	16 GB	160 GB	160 GB	0 GB	Yes

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Age

3 days, 2 hours

Create Snapshot

3 days, 20 hours

Create Snapshot

Cancel

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Networks provide the communication channels for instances in the cloud. You can select ports instead of networks or a mix of both.

Allocated

Displaying 0 items

Network	Subnets Associated	Shared	Admin State	Status
Select one or more networks from the available networks below.				
Displaying 0 items				

Available

3

Select one or more

Click here for filters or full text search.

Displaying 3 items

Network	Subnets Associated	Shared	Admin State	Status
external124	external-subnet124	No	Up	Active
shared	shared-subnet	No	Up	Active
private	private-subnet ipv6-private-subnet	No	Up	Active

Displaying 3 items

Cancel

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Metadata

Networks provide the communication channels for instances in the cloud. You can select ports instead of networks or a mix of both.

Allocated

1

Displaying 1 item

Network	Subnets Associated	Shared	Admin State	Status
external124	external-subnet124	No	Up	Active

Displaying 1 item

Available

2

Select one or more

Click here for filters or full text search.

Displaying 2 items

Network	Subnets Associated	Shared	Admin State	Status
shared	shared-subnet	No	Up	Active
private	private-subnet ipv6-private-subnet	No	Up	Active

Displaying 2 items

Cancel

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Instance ID = Filter Launch Instance Delete Instances More Actions

Displaying 3 items

	Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
☐	rhel-demo02	-	10.200.124.193	ds2G	test01	Build	nova	Block Device Mapping	No State	0 minutes	Associate Floating IP
☐	01rhel95	-	10.200.124.200	m1.medium	test01	Active	nova	None	Running	3 days, 2 hours	Create Snapshot
☐	test-vm124-ssh	ubuntu-jammy	10.200.124.191	m1.tiny	test01	Active	nova	None	Running	3 days, 20 hours	Create Snapshot

Displaying 3 items

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Instance ID = Filter Launch Instance Delete Instances More Actions

Displaying 3 items

	Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
☐	rhel-demo02	-	10.200.124.193	ds2G	test01	Active	nova	None	Running	0 minutes	Create Snapshot
☐	01rhel95	-	10.200.124.200	m1.medium	test01	Active	nova	None	Running	3 days, 2 hours	Create Snapshot
☐	test-vm124-ssh	ubuntu-jammy	10.200.124.191	m1.tiny	test01	Active	nova	None	Running	3 days, 20 hours	Create Snapshot

Displaying 3 items

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Instance ID = Filter Launch Instance Delete Instances More Actions

Displaying 3 items

☐	Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
☐	rhel-demo02	-	10.200.124.193	ds2G	test01	Active	nova	None	Running	0 minutes	Create Snapshot
☐	01rhel95	-	10.200.124.200	m1.medium	test01	Active	nova	None	Running	3 days, 2 hou	Associate Floating IP Attach Interface Detach Interface Edit Instance Attach Volume Detach Volume Update Metadata Edit Security Groups Edit Port Security Groups Console View Log Rescue Instance Pause Instance Suspend Instance Shelve Instance Resize Instance Lock Instance
☐	test-vm124-ssh	ubuntu-jammy	10.200.124.191	m1.tiny	test01	Active	nova	None	Running	3 days, 20 hc	

Displaying 3 items

http://10.200.124.188/dashboard/project/instances/#

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rhel-demo02

Overview Interfaces Log Console Action Log

Instance Console

If console is not responding to keyboard input: click the grey status bar below. [Click here to show only console](#)
To exit the fullscreen mode, click the browser's back button.

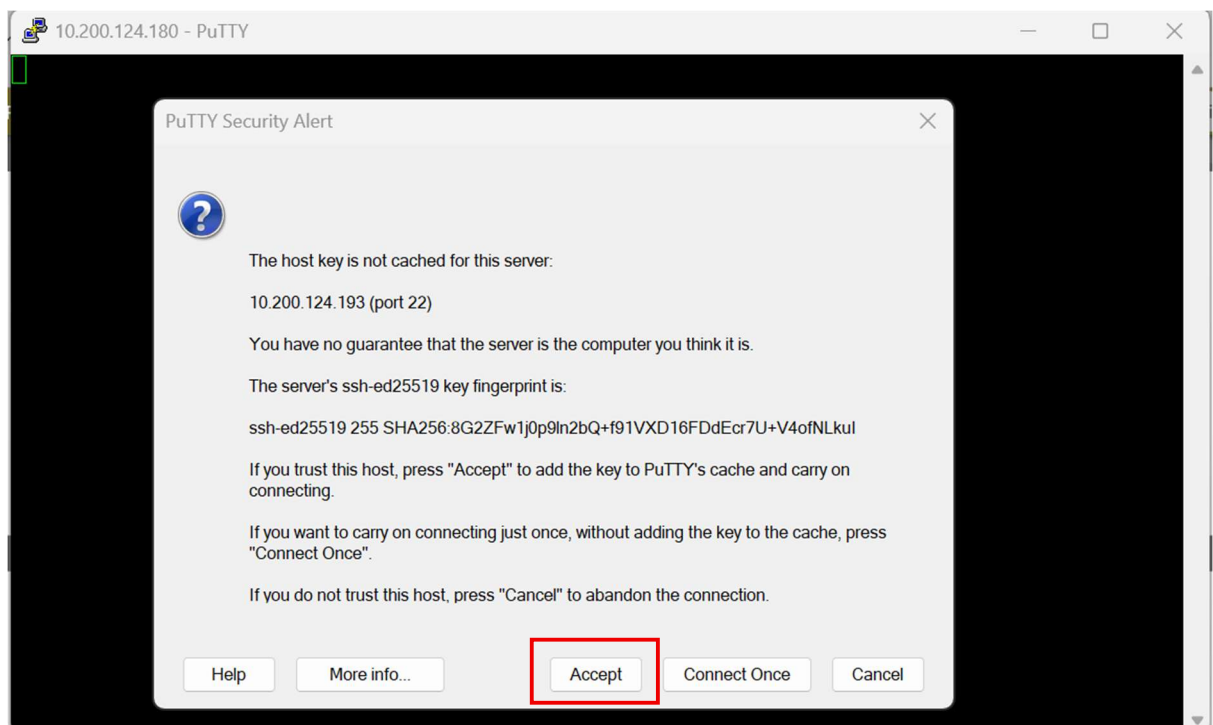
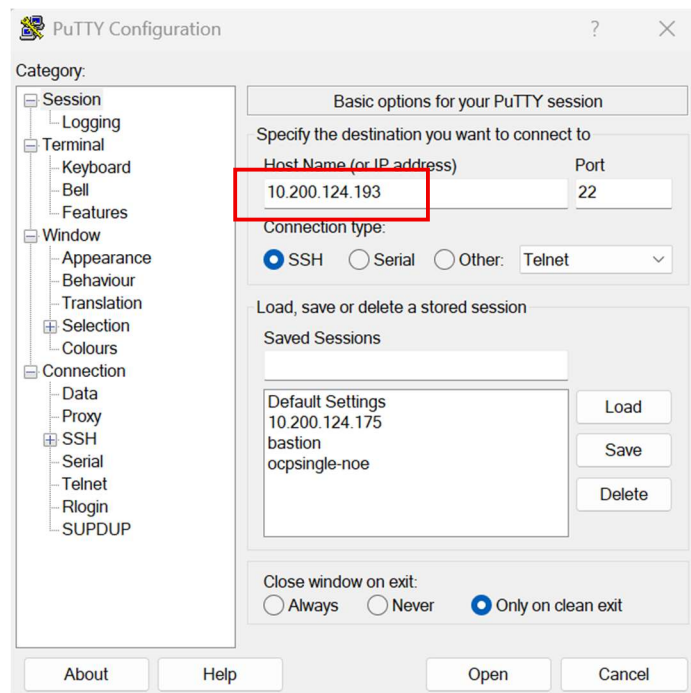
Connected to QEMU (instance-00000010) Send Ctrl+Alt+Del

Red Hat Enterprise Linux 9.5 (Plow)
Kernel 5.14.0-583.11.1.el9_5.x86_64 on an x86_64
rhel-demo01 login: _

```
Red Hat Enterprise Linux 9.5 (Plow)
Kernel 5.14.0-503.11.1.el9_5.x86_64 on an x86_64

rhel-demo01 login: root
Password:
[root@rhel-demo01 ~]# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether fa:16:3e:10:65:6a brd ff:ff:ff:ff:ff:ff
    altname ens3-0
    inet 10.200.124.193/23 brd 10.200.125.255 scope global dynamic noprefixroute ens3
        valid_lft 42954sec preferred_lft 42954sec
    inet6 fe80::f816:3eff:fe10:656a/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
[root@rhel-demo01 ~]#
```

1.3 ทดสอบเข้าถึง Instance โดยการ remote (SSH)



```
root@rhel-demo01:~  
login as: root  
root@10.200.124.193's password:  
Register this system with Red Hat Insights: rhc connect  
  
Example:  
# rhc connect --activation-key <key> --organization <org>  
  
The rhc client and Red Hat Insights will enable analytics and additional  
management capabilities on your system.  
View your connected systems at https://console.redhat.com/insights  
  
You can learn more about how to register your system  
using rhc at https://red.ht/registration  
Last login: Fri Aug 28 11:38:49 2026 from 192.168.191.167  
[root@rhel-demo01 ~]# ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host  
        valid_lft forever preferred_lft forever  
2: ens3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether fa:16:3e:10:65:6a brd ff:ff:ff:ff:ff:ff  
    altname enp0s3  
    inet 10.200.124.193/23 brd 10.200.125.255 scope global dynamic noprefixroute ens3  
        valid_lft 42752sec preferred_lft 42752sec  
    inet6 fe80::f816:3eff:fe10:656a/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
[root@rhel-demo01 ~]#
```

2. Create new instance by QCOW2 and set SSH Keypair

2.1 Upload Image

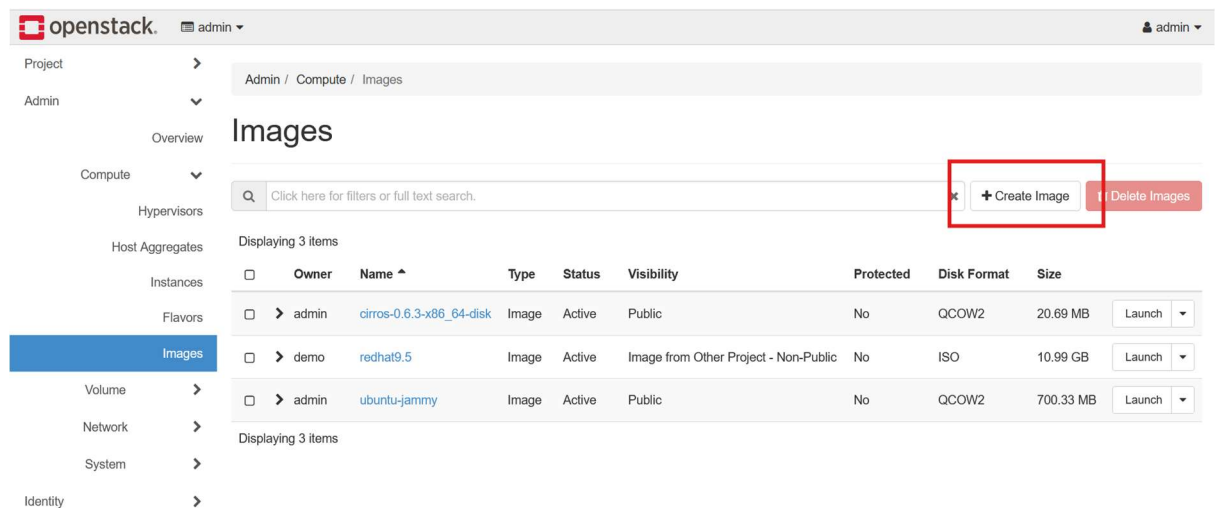
Upload imageโดย Command Line

```
source ~/devstack/openrc admin admin wget https://cloud-images.ubuntu.com/jammy/current/jammy-server-cloudimg-amd64.img  
  
openstack image create --disk-format qcow2 --container-format bare --public --file jammy-server-cloudimg-amd64.img ubuntu-jammy
```

ตรวจสอบ Image

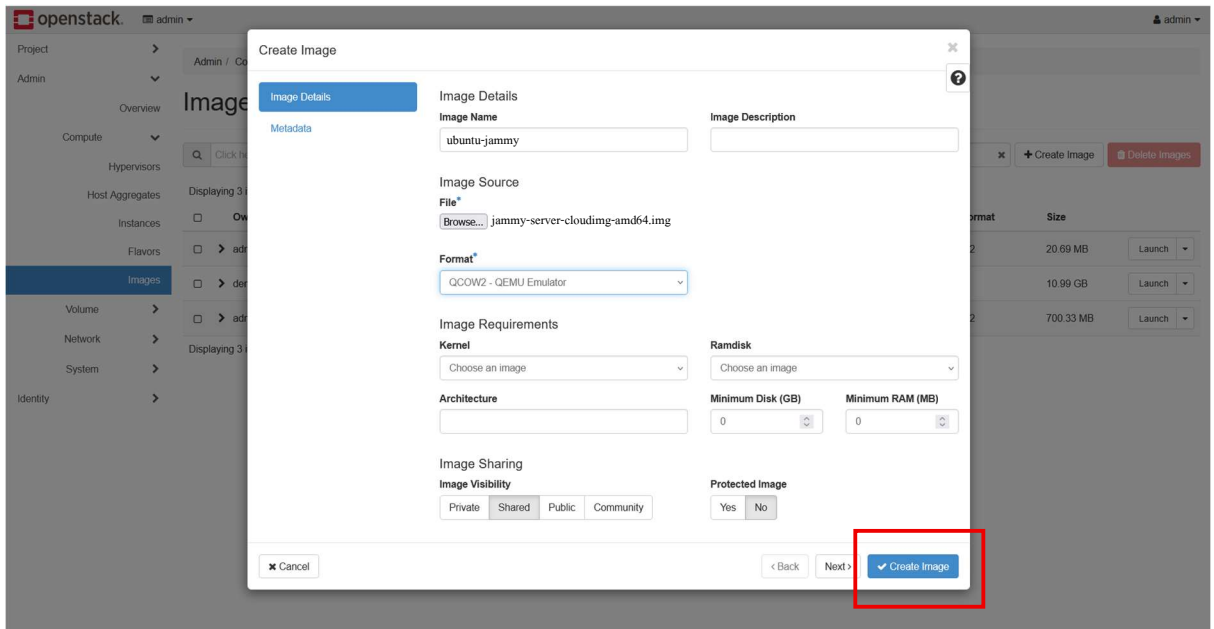
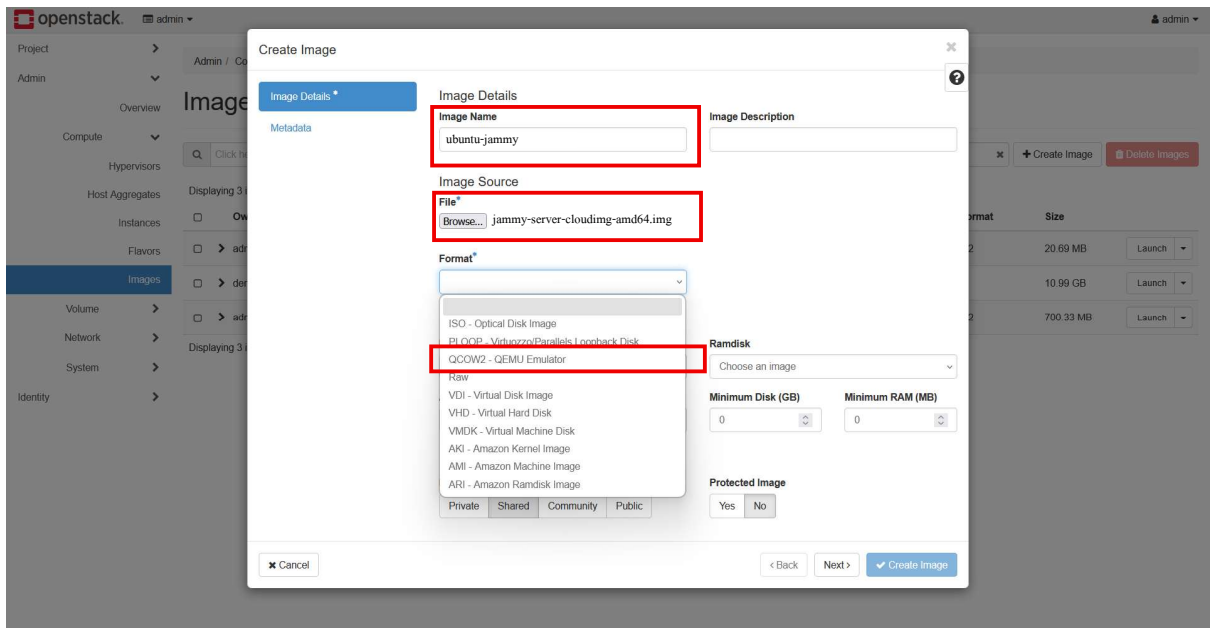
```
openstack image list
```

Upload Imageโดยหน้า Web Console

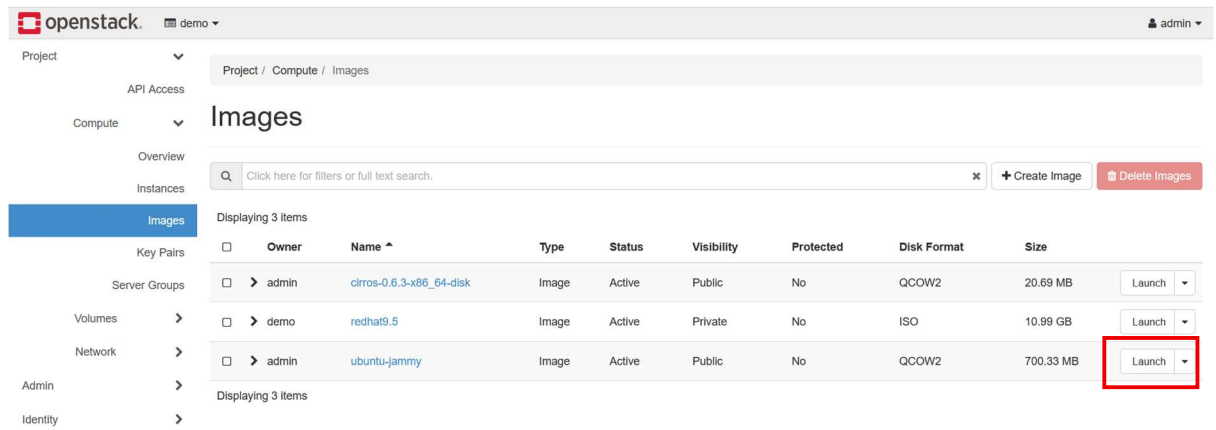


The screenshot shows the OpenStack Web Console interface. The left sidebar contains navigation links: Project, Admin, Overview, Compute, Hypervisors, Host Aggregates, Instances, Flavors, Images (highlighted), Volume, Network, System, and Identity. The main content area is titled 'Images' and displays a table of existing images. A red box highlights the '+ Create Image' button in the top right corner of the main content area.

	Owner	Name	Type	Status	Visibility	Protected	Disk Format	Size	
☐	admin	cirros-0.6.3-x86_64-disk	Image	Active	Public	No	QCOW2	20.69 MB	Launch
☐	demo	redhat9.5	Image	Active	Image from Other Project - Non-Public	No	ISO	10.99 GB	Launch
☐	admin	ubuntu-jammy	Image	Active	Public	No	QCOW2	700.33 MB	Launch



2.2 Create Instance



openstack demo admin

Project / Compute / Images

Images

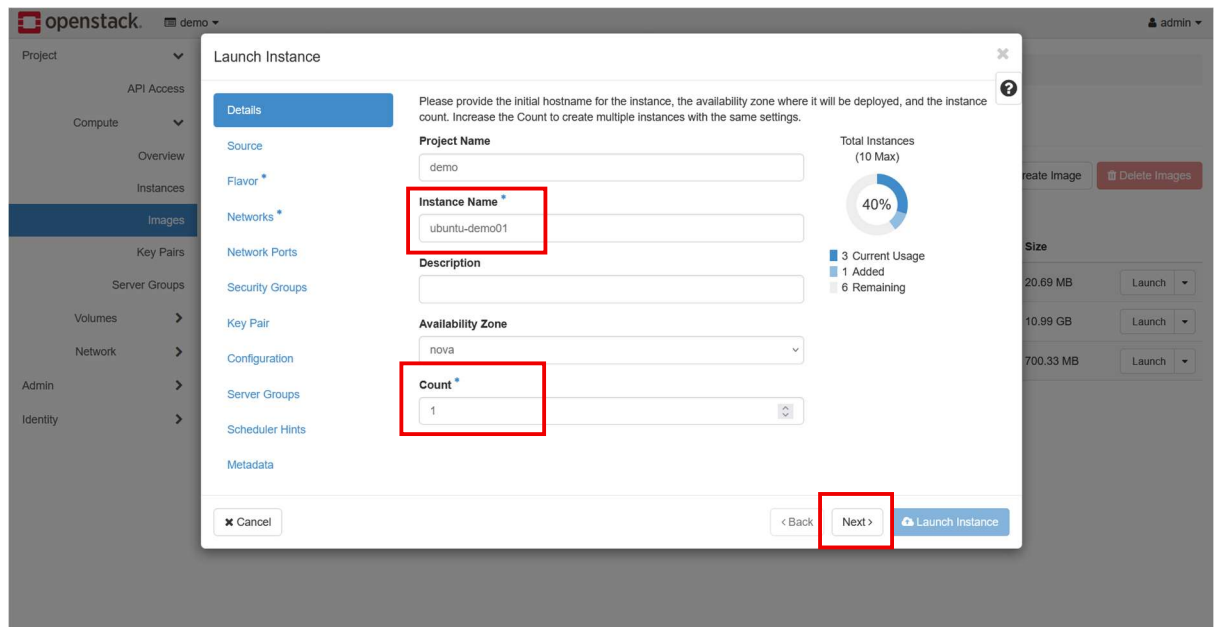
Click here for filters or full text search.

+ Create Image Delete Images

Displaying 3 Items

	Owner	Name ^	Type	Status	Visibility	Protected	Disk Format	Size	
☐	admin	cirros-0.6.3-x86_64-disk	Image	Active	Public	No	QCOW2	20.69 MB	Launch
☐	demo	redhat9.5	Image	Active	Private	No	ISO	10.99 GB	Launch
☐	admin	ubuntu-jammy	Image	Active	Public	No	QCOW2	700.33 MB	Launch

Displaying 3 Items



openstack demo admin

Project / Compute / Images

Launch Instance

Please provide the initial hostname for the instance, the availability zone where it will be deployed, and the instance count. Increase the Count to create multiple instances with the same settings.

Project Name: demo

Instance Name: [ubuntu-demo01](#)

Description:

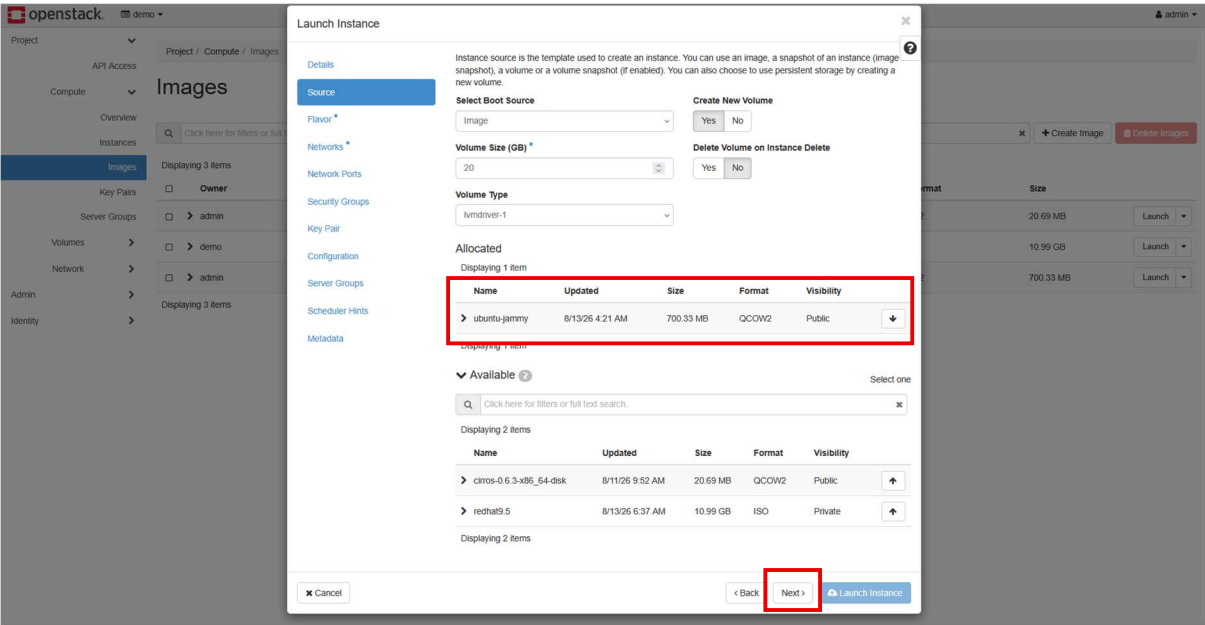
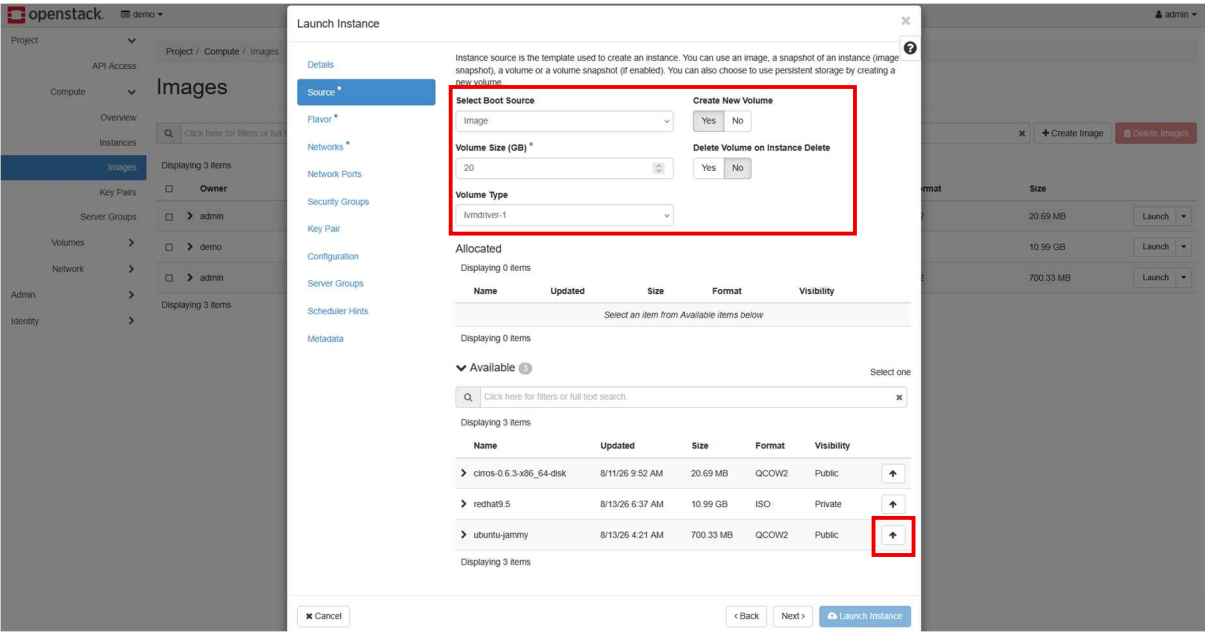
Availability Zone: nova

Count: 1

Total Instances (10 Max): 40%

3 Current Usage, 1 Added, 6 Remaining

< Back Next > Launch Instance



openstackdemo

Project / Compute / Images

Images

Displaying 3 items

Owner

admin

demo

admin

Displaying 3 items

Flavor

Networks

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Select a flavor from the available flavors below.

Available12

Click here for filters or full text search.

Displaying 12 items

Name	vCPUs	RAM	Total Disk	Root Disk	Ephemeral Disk	Public
m1.nano	1	192 MB	1 GB	1 GB	0 GB	Yes
m1.micro	1	256 MB	1 GB	1 GB	0 GB	Yes
cirros256	1	256 MB	1 GB	1 GB	0 GB	Yes
m1.tiny	1	512 MB	1 GB	1 GB	0 GB	Yes
ds512M	1	512 MB	5 GB	5 GB	0 GB	Yes
ds1G	1	1 GB	10 GB	10 GB	0 GB	Yes
m1.small	1	2 GB	20 GB	20 GB	0 GB	Yes
ds2G	2	2 GB	10 GB	10 GB	0 GB	Yes
m1.medium	2	4 GB	40 GB	40 GB	0 GB	Yes
ds4G	4	4 GB	20 GB	20 GB	0 GB	Yes
m1.large	4	8 GB	80 GB	80 GB	0 GB	Yes
m1.xlarge	8	16 GB	160 GB	160 GB	0 GB	Yes

Displaying 12 items

CancelBackNextLaunch Instance

admin

Create ImageDelete Images

FormatSize

20.69 MBLaunch

10.99 GBLaunch

700.33 MBLaunch

openstackdemo

Project / Compute / Images

Images

Displaying 3 items

Owner

admin

demo

admin

Displaying 3 items

Flavor

Networks

Network Ports

Security Groups

Key Pair

Configuration

Server Groups

Scheduler Hints

Metadata

Source

Displaying 1 item

Name	vCPUs	RAM	Total Disk	Root Disk	Ephemeral Disk	Public
m1.small	1	2 GB	20 GB	20 GB	0 GB	Yes

Displaying 1 item

Available11

Click here for filters or full text search.

Displaying 11 items

Name	vCPUs	RAM	Total Disk	Root Disk	Ephemeral Disk	Public
m1.nano	1	192 MB	1 GB	1 GB	0 GB	Yes
m1.micro	1	256 MB	1 GB	1 GB	0 GB	Yes
cirros256	1	256 MB	1 GB	1 GB	0 GB	Yes
m1.tiny	1	512 MB	1 GB	1 GB	0 GB	Yes
ds512M	1	512 MB	5 GB	5 GB	0 GB	Yes
ds1G	1	1 GB	10 GB	10 GB	0 GB	Yes
ds2G	2	2 GB	10 GB	10 GB	0 GB	Yes
m1.medium	2	4 GB	40 GB	40 GB	0 GB	Yes
ds4G	4	4 GB	20 GB	20 GB	0 GB	Yes
m1.large	4	8 GB	80 GB	80 GB	0 GB	Yes
m1.xlarge	8	16 GB	160 GB	160 GB	0 GB	Yes

Displaying 11 items

CancelBackNextLaunch Instance

admin

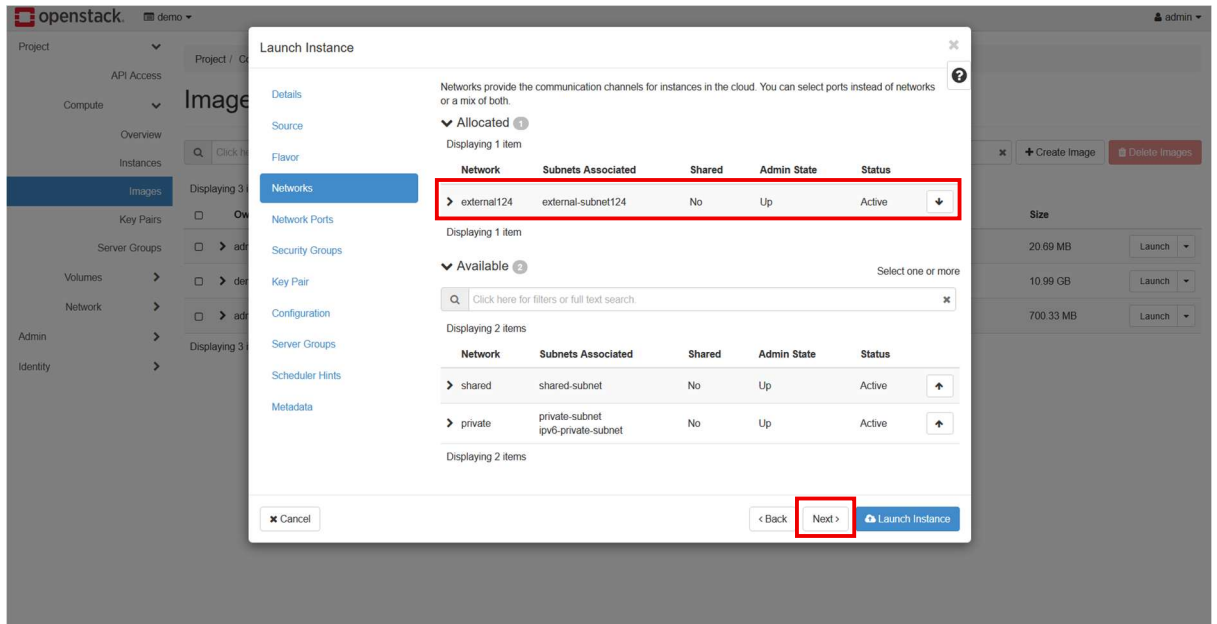
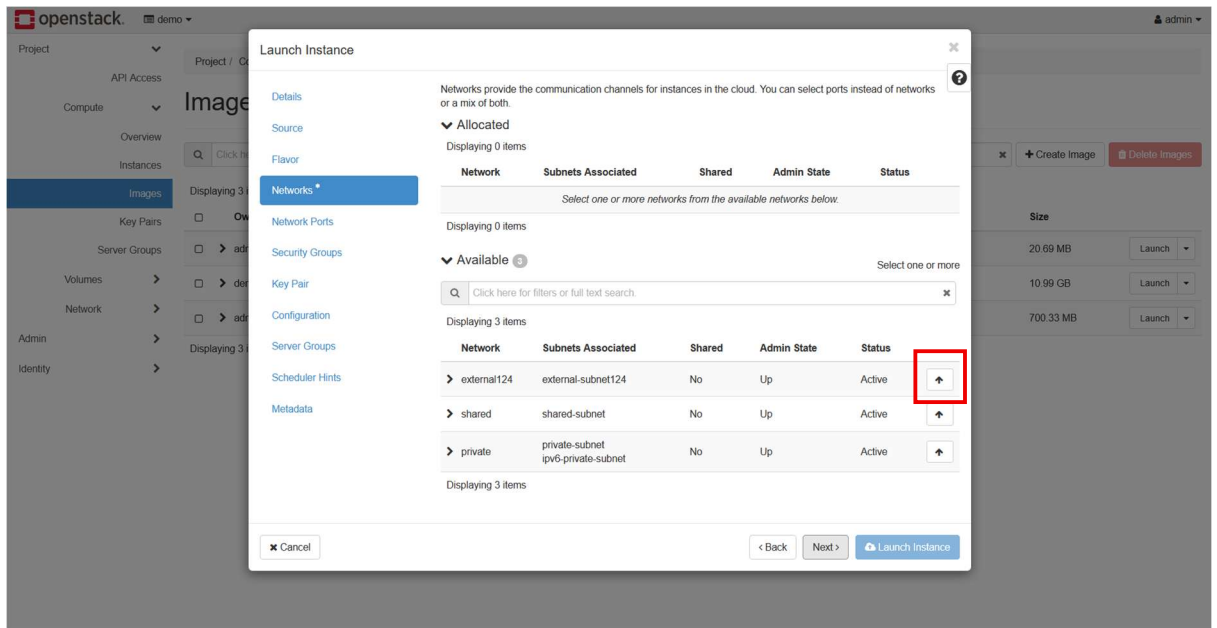
Create ImageDelete Images

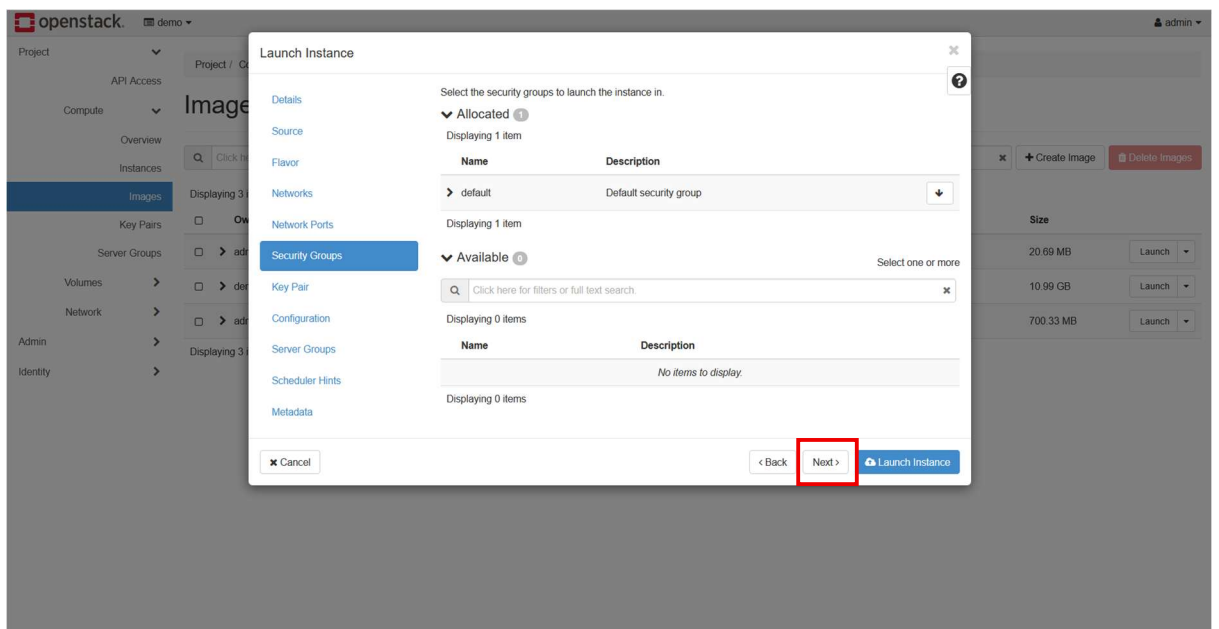
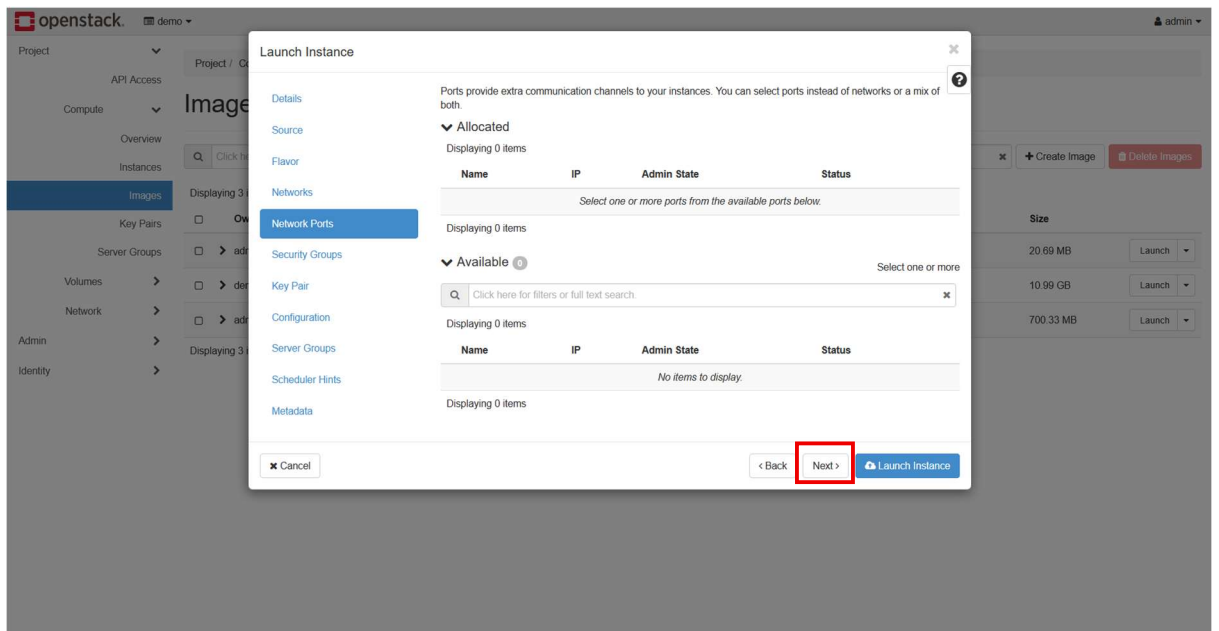
FormatSize

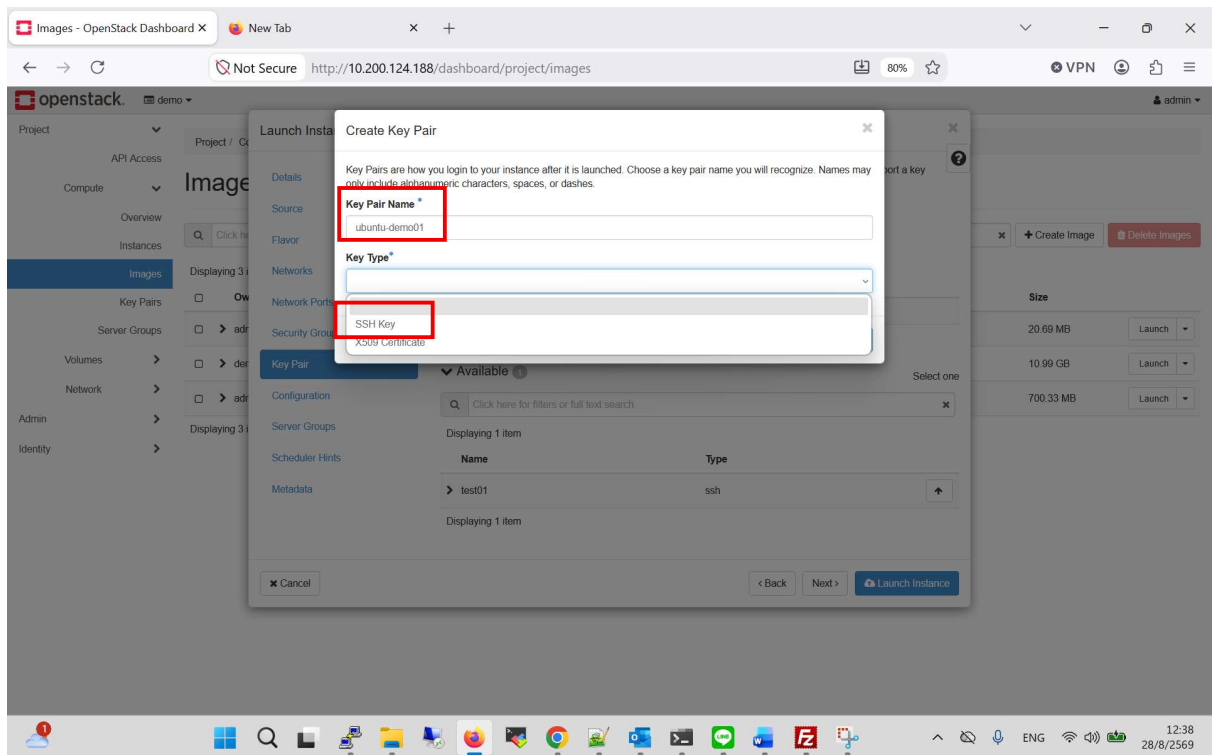
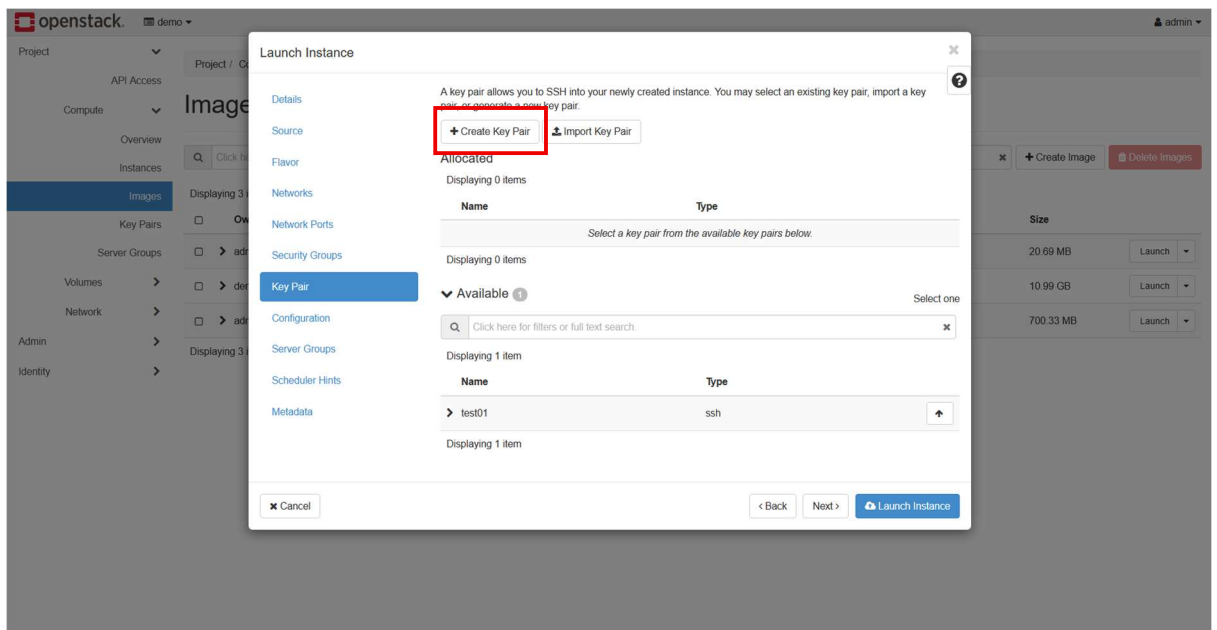
20.69 MBLaunch

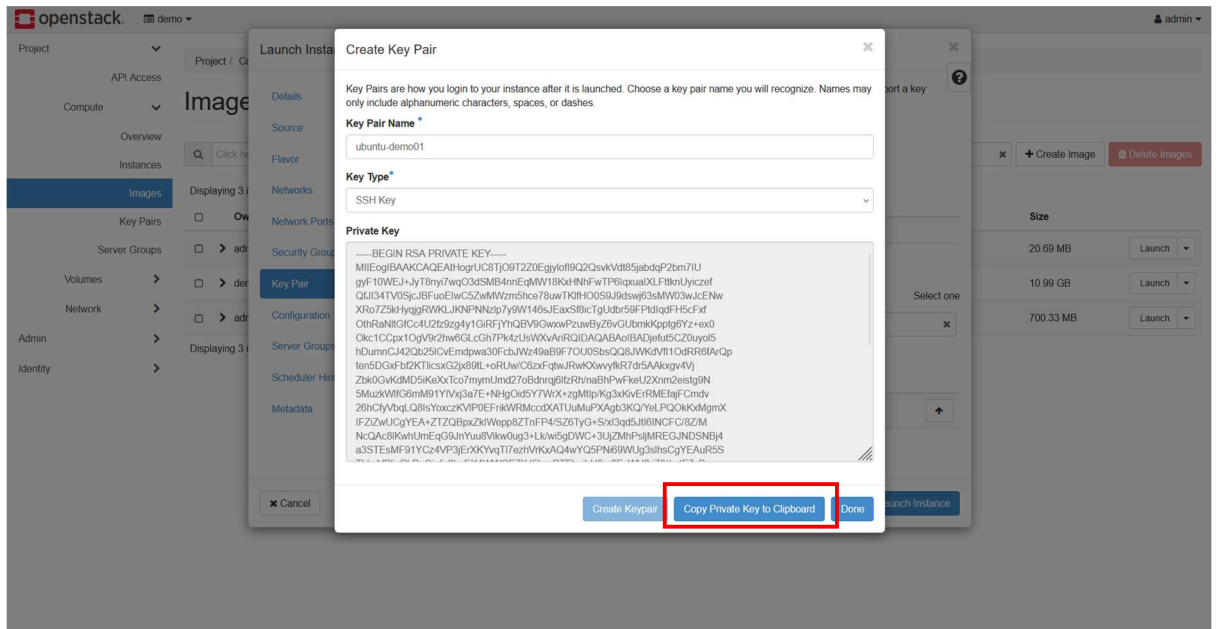
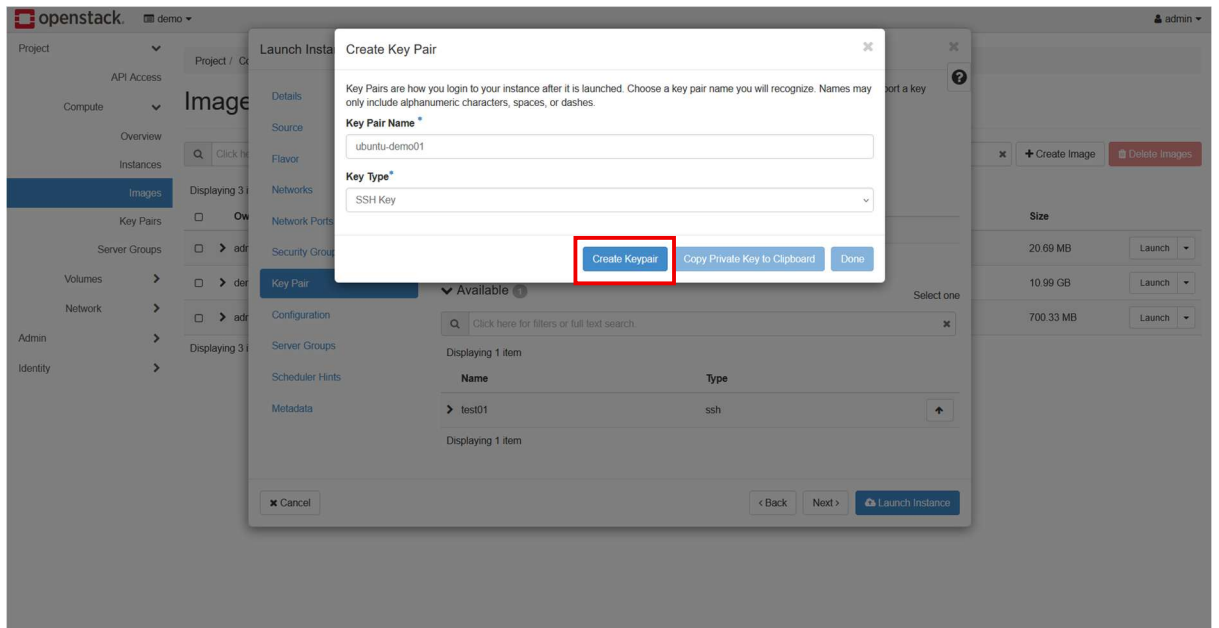
10.99 GBLaunch

700.33 MBLaunch

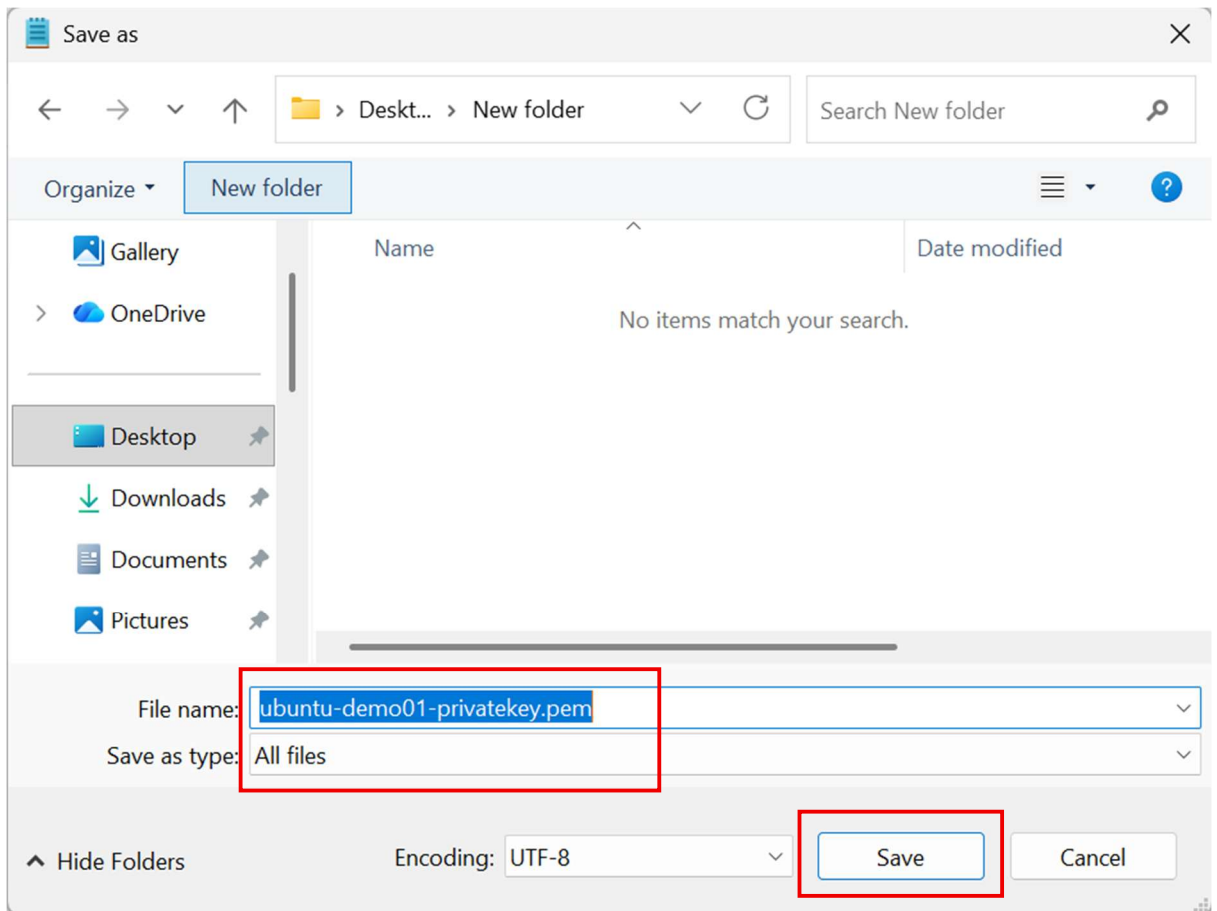


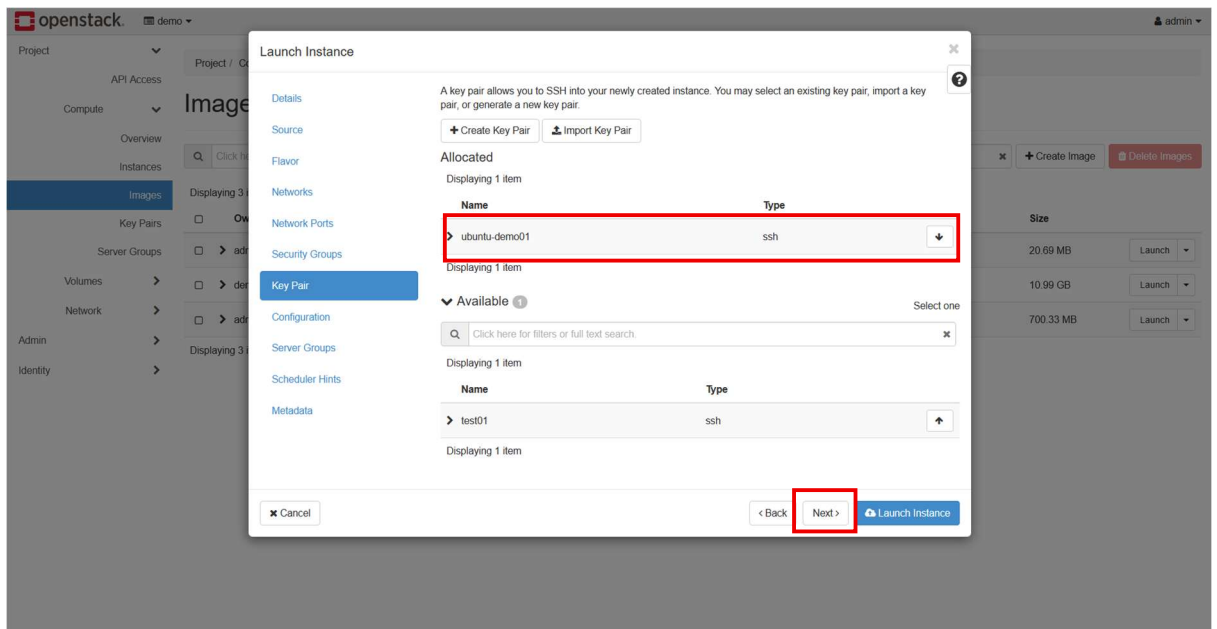
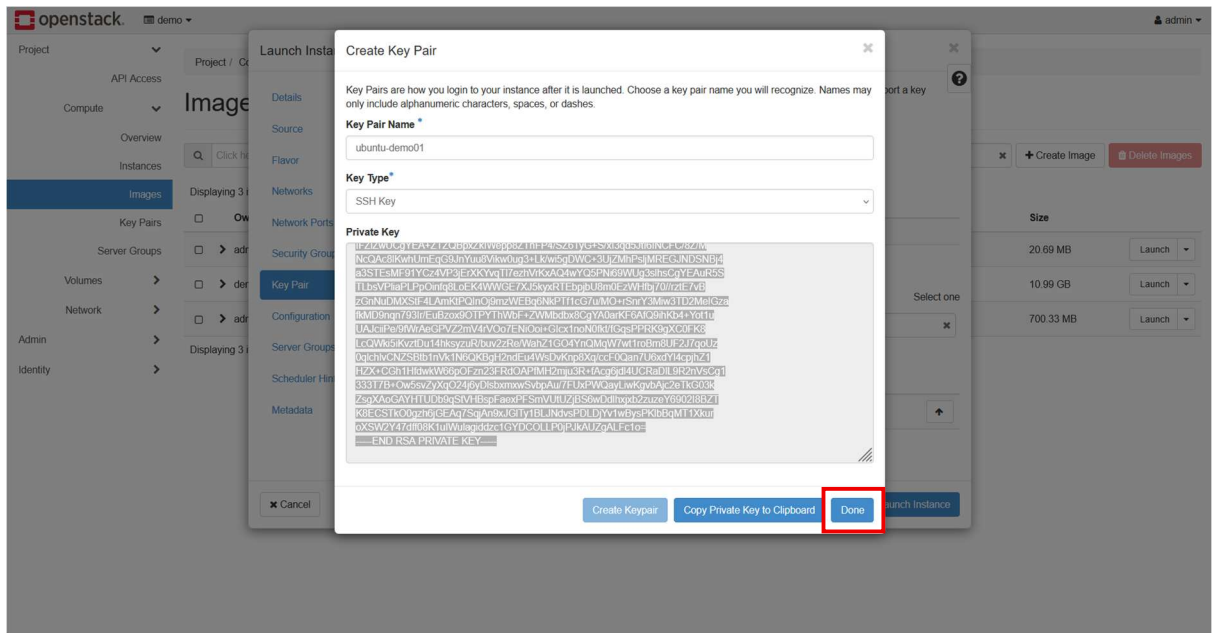


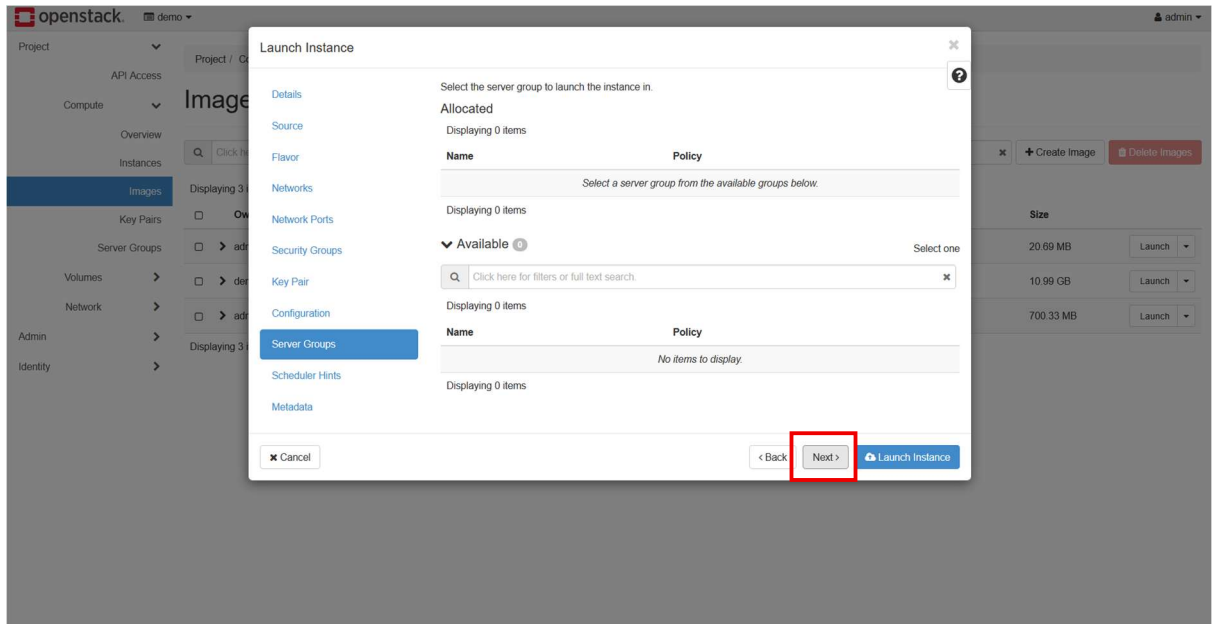
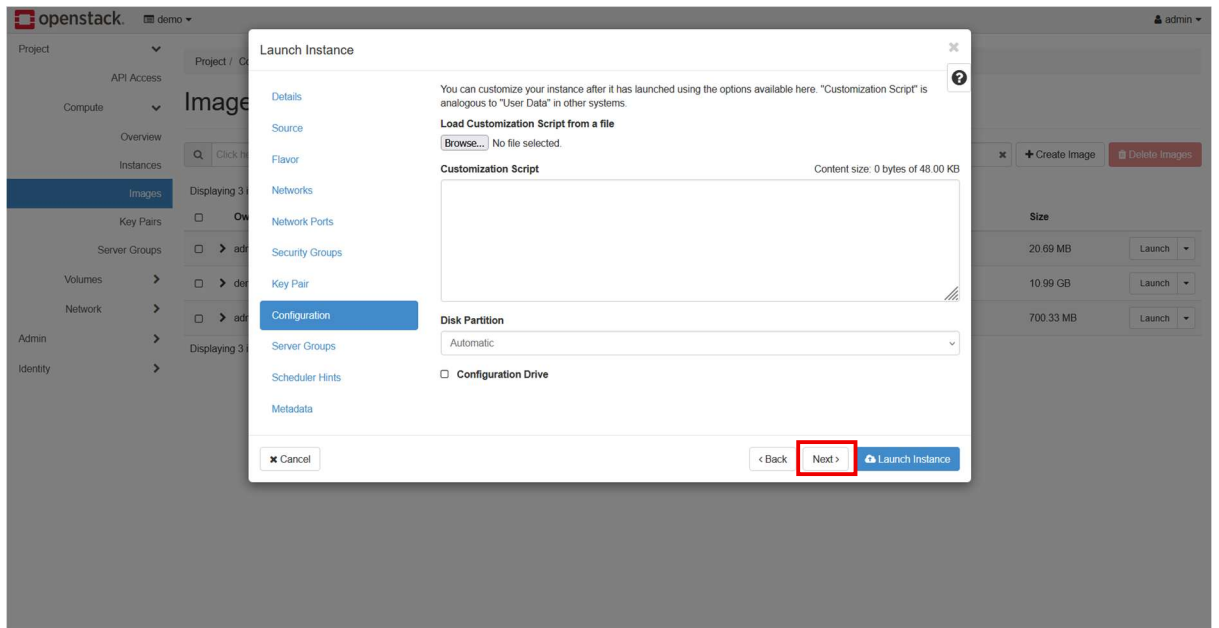


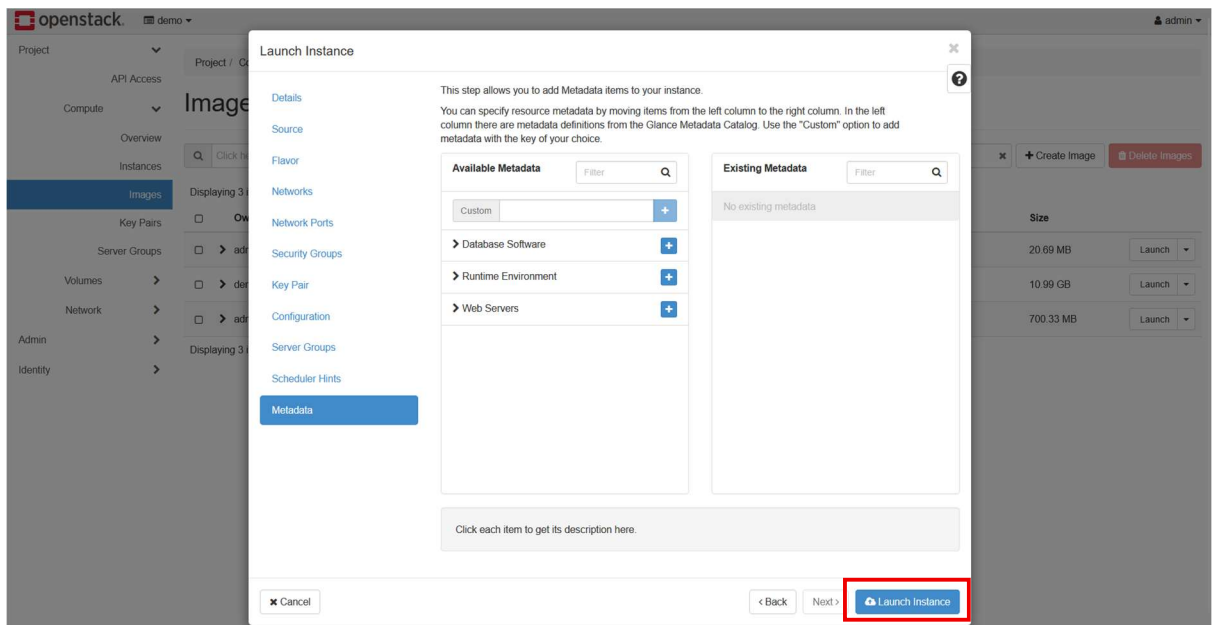
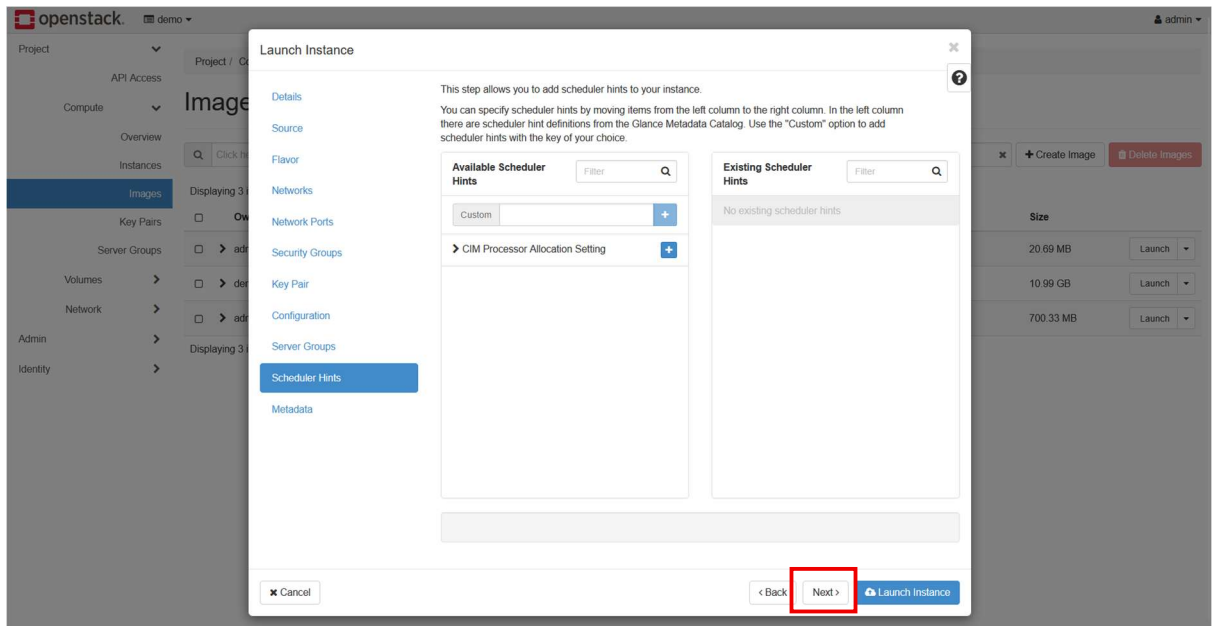


```
-----BEGIN RSA PRIVATE KEY-----
MIIEpQIBAAKCAQEAJwC5ZwMwZm5hce78uWTK1fH00S9J9dswj63sMW03WjCENw
XR07Z5kHyqjgRWKLJKPNnzlp7y9W146sJEaxSf8icTgUdbr59FPtdIqdFH5cFxf
OthRanltGfCc4U2fz9zg4y1GiRFjYhQBv9GwxwPzuwByZ6vGUbmKkpptg6Yz+ex0
Okc1CCpx10gV9r2hw6GLcGh7Pk4zUsWxvArIRQIDAQAABADjefut5CZ0uyo15
hDumncJ42Qb251CvEmdpwa30FcbJWz49aB9F70U0SbsQQ8JWkdVFI10dRR6fArQp
ten5DGxvFb2KtlicsXG2jx89tL+oRUw/C6zxFqtWJRwKXwvyfKR7dr5AAkxgv4Vj
Zbk0GvKdMD5iKeXxTco7mymUmd27oBdnrqj61fzRh/naBhPwFkeU2Xnm2eistg9N
5Muzkwl1f66mM91YIVxj3a7E+NHgOid5Y7WrX+zgMtIp/Kg3xKivErRMEfajFCmdv
26hCfyVbQLO8IsYoxczKV1P0EFrikWRMccdxATUuMuPXAgb3KQ/YeLPQOkKxMgmX
IFZiZwUCgYEA+ZTQ8pxZkIwepp8ZTnFP4/SZ6TyG+S/xI3qd5JtI6INCFC/8Z/M
NcQA81KwhUmEqG9JnYuu8Vikw0ug3+Lk/wi5gDWC+3UjZMhPs1jMREGJNDSNBj4
a3STESMF91YCz4VP3jErXKYvqT17ezhVrKxQA4wYQ5PNi69WUg3s1hsCgYEAuR5S
TLbsVP1iaPLPp0infq8LoEK4WwGE7XJ5kyxRTEbpjbU8m0EzWHfbj70//rztE7vB
zGnNuDMXStf4LAmKtPQInOj9mzWEBq6NkPTf1cG7u/MO+rSnY3Miw3TD2MeIGza
fkMD9nqn793Ir/EuBzox90TPYThWbF+ZWmbdbx8CgYA0arKF6AfQ9ihKb4+Yot1u
UAJciPe/9fWraeGPVZ2mV4rV0o7ENi0oi+Gicx1noN0fkt/fGqsPPRK9gXC0FK8
LcQWki5iKvztDu14hksyzuR/buv2zRe/WahZ1G04YnQMqW7wt1roBm8UF2J7qoUz
0qlchlVbCNZSBtb1nVkiN6QKBGH2ndEu4WsDvKnp8Xq/ccF0Qan7U6xdY14cpjhZ1
HZX+CGh1HfdwkW66pOFzn23FRd0APfMH2mju3R+fAcg6jd14UCRaD1L9R2nVsCg1
333T7B+0w5svZyXq024j6yD1sbxmwSvbpAu/7FuxPWQayLiWkvbAjc2eTkG03k
ZsgXAoGAYHTUDb9qStVHBspFaexPFsmVUtUZjBS6wDd1hxjxb2zuzeY6902I8BZT
K8ECSTK00gzh6jGEAq7SqjAn9xJG1Ty1BLJNdvSPDLjYv1wBysPK1bBqMT1Xkur
oXSW2Y4df08K1uIWulagiddzc1GYDCOLLp0jPjKAUZgALFc1o=
-----END RSA PRIVATE KEY-----
```









Project / Compute / Instances

Instances

Instance ID ▾ Filter Launch Instance Delete Instances More Actions ▾

Displaying 4 items

☐	Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
☐	ubuntu-demo01	-	10.200.124.202	m1.small	ubuntu-demo01	Active	eu-nova	None	Running	0 minutes	Create Snapshot
☐	rhel-demo02	-	10.200.124.193	ds2G	test01	Active	eu-nova	None	Running	2 hours, 23 minutes	Create Snapshot
☐	01rhel95	-	10.200.124.200	m1.medium	test01	Shutoff	eu-nova	None	Shut Down	3 days, 5 hours	Start Instance
☐	test-vm124-ssh	ubuntu-jammy	10.200.124.191	m1.tiny	test01	Shutoff	eu-nova	None	Shut Down	3 days, 23 hours	Start Instance

Displaying 4 items

http://10.200.124.188/dashboard/project/api_access/

2.3 ทดสอบเข้าถึง Instance โดยการ remote (SSH)

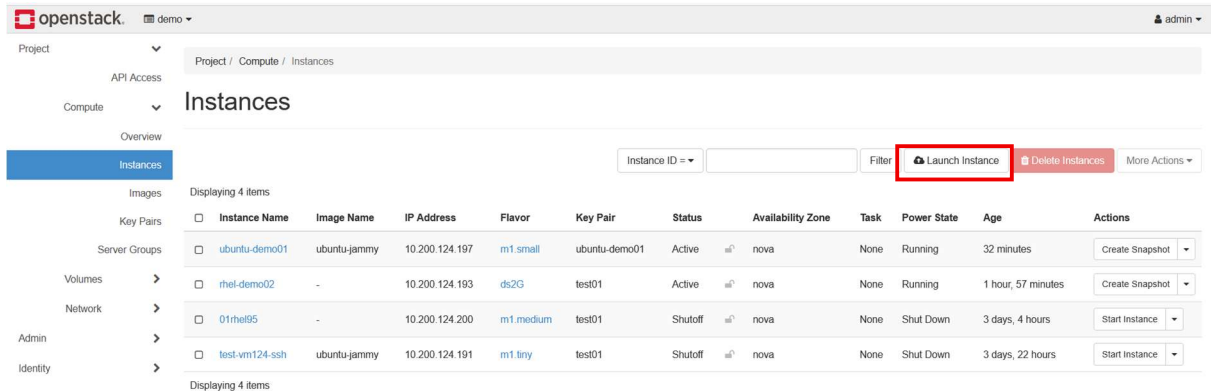
```
ssh -i ubuntu-demo01-privatekey.pem ubuntu@10.200.124.202
```

```
3. ubuntu@ubuntu-demo01: ~  
28/08/2026 13:58:17 /cygdrive/c/Users/nawindet/Desktop/New folder ssh -i ubuntu-demo01-privatekey.pem ubuntu@10.200.124.202  
Warning: Permanently added '10.200.124.202' (ECDSA) to the list of known hosts.  
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-187-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Fri Aug 28 06:58:36 UTC 2026  
  
System load: 0.15          Processes:           90  
Usage of /:  8.4% of 19.20GB Users logged in:     0  
Memory usage: 9%          IPv4 address for ens3: 10.200.124.202  
Swap usage:  0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
/usr/bin/xdm: file /home/ubuntu/.Xauthority does not exist  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.
```

วิธีนี้จะไม่สามารถ login ผ่านหน้า console ผ่าน user password ได้

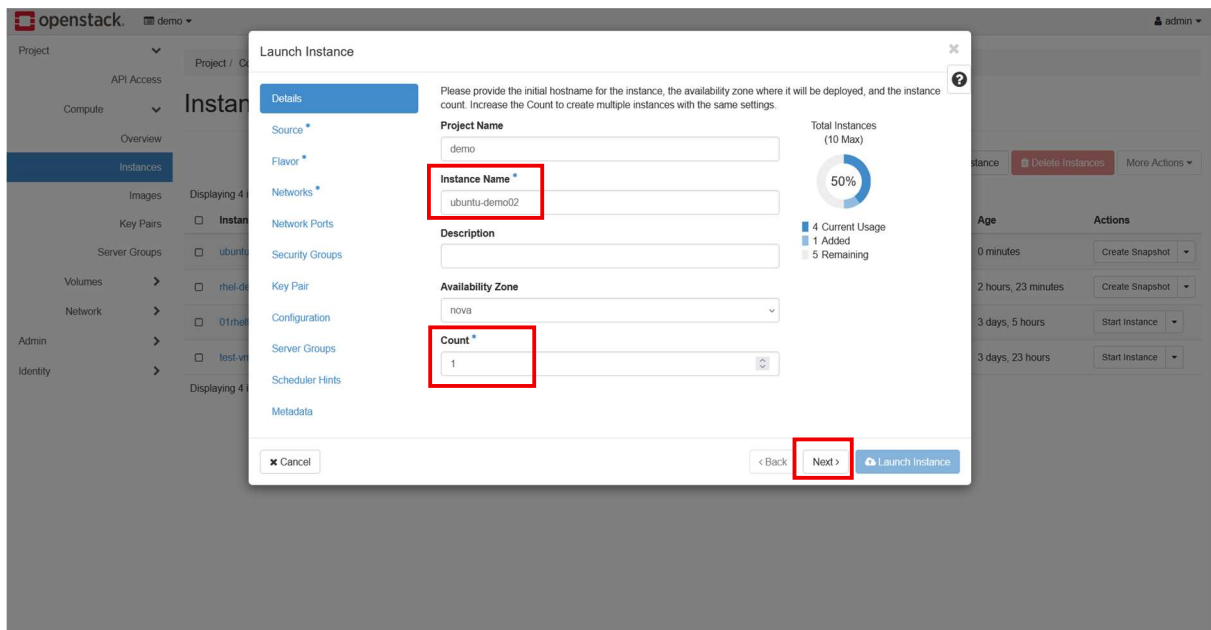
3. Create new instance by QCOW2 and set Cloud init script

3.1 Create instance

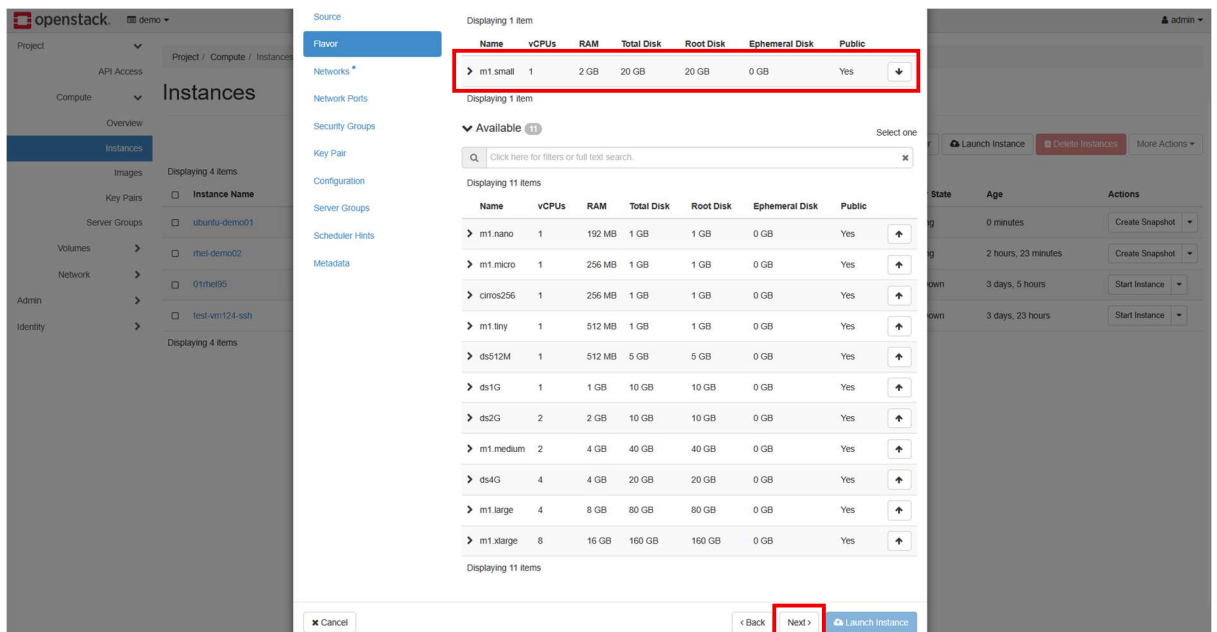
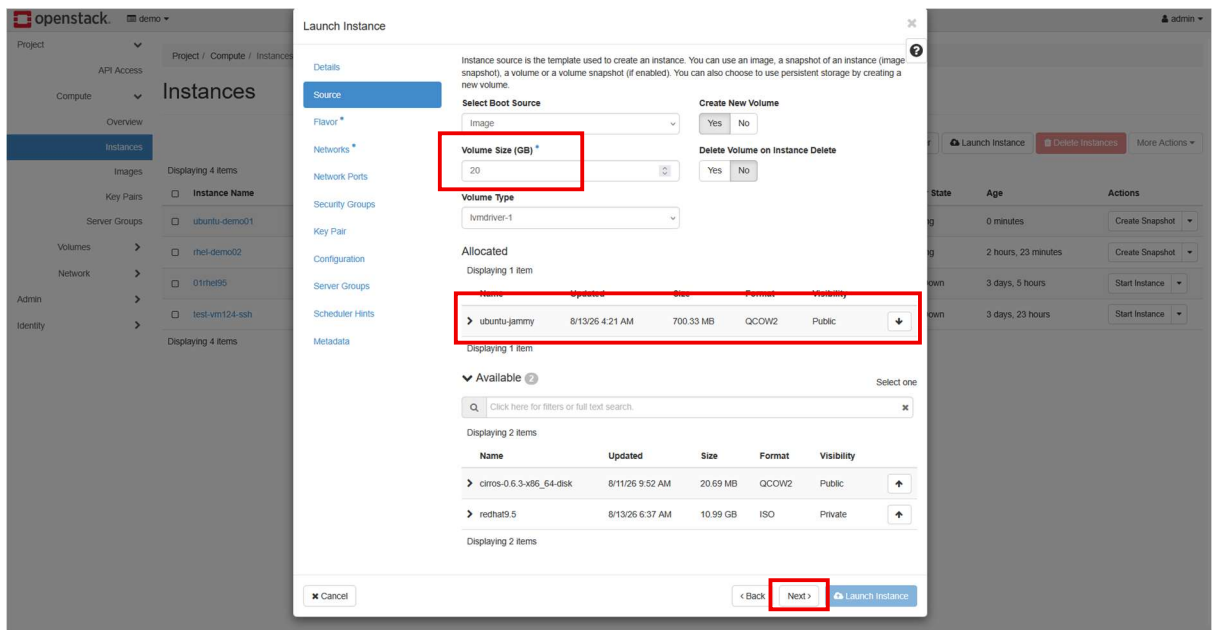


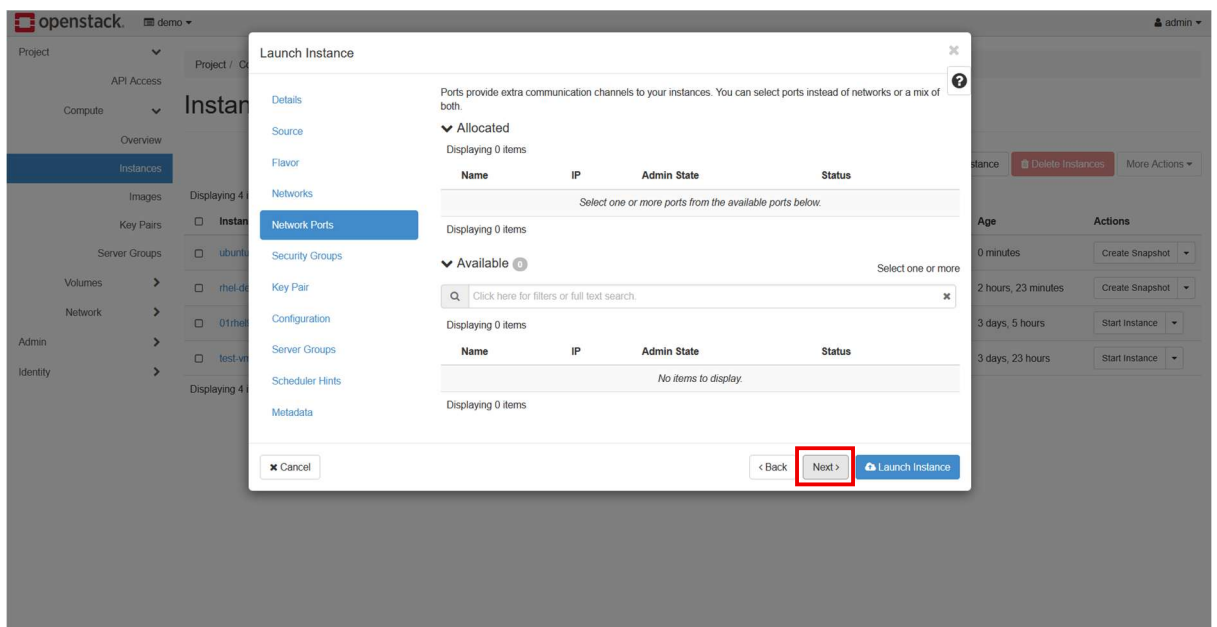
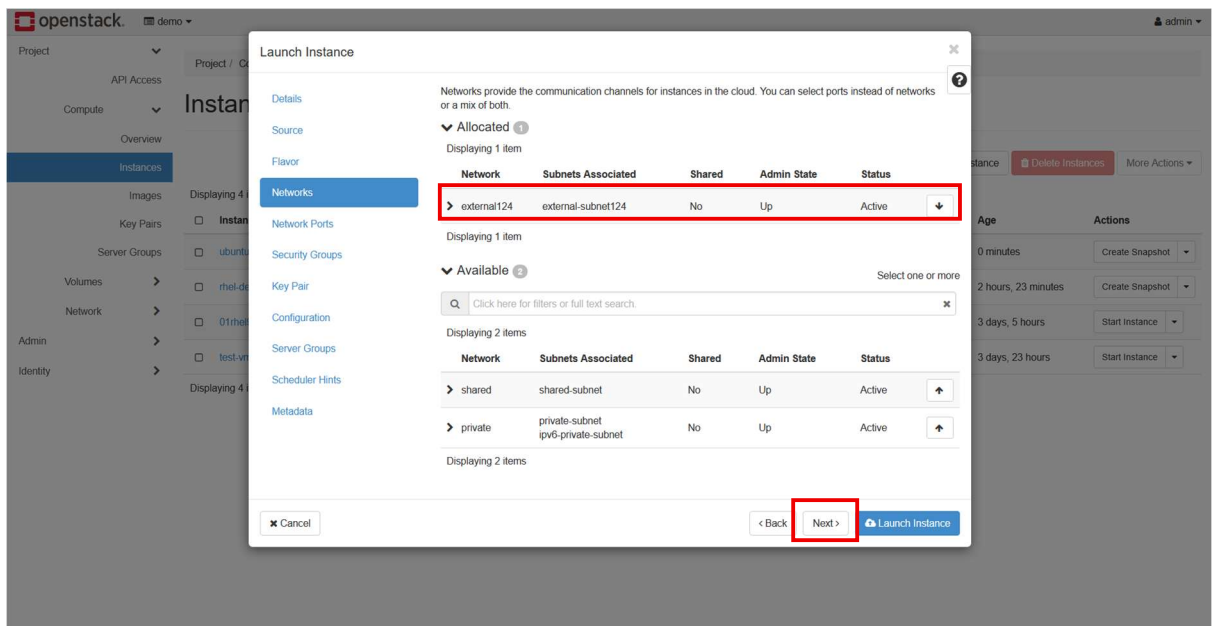
The screenshot shows the OpenStack dashboard interface. On the left is a navigation sidebar with categories like Project, API Access, Compute, Overview, and Instances (which is selected). The main content area is titled 'Instances' and shows a table of existing instances. Above the table, there are filters and action buttons. The 'Launch Instance' button is highlighted with a red box.

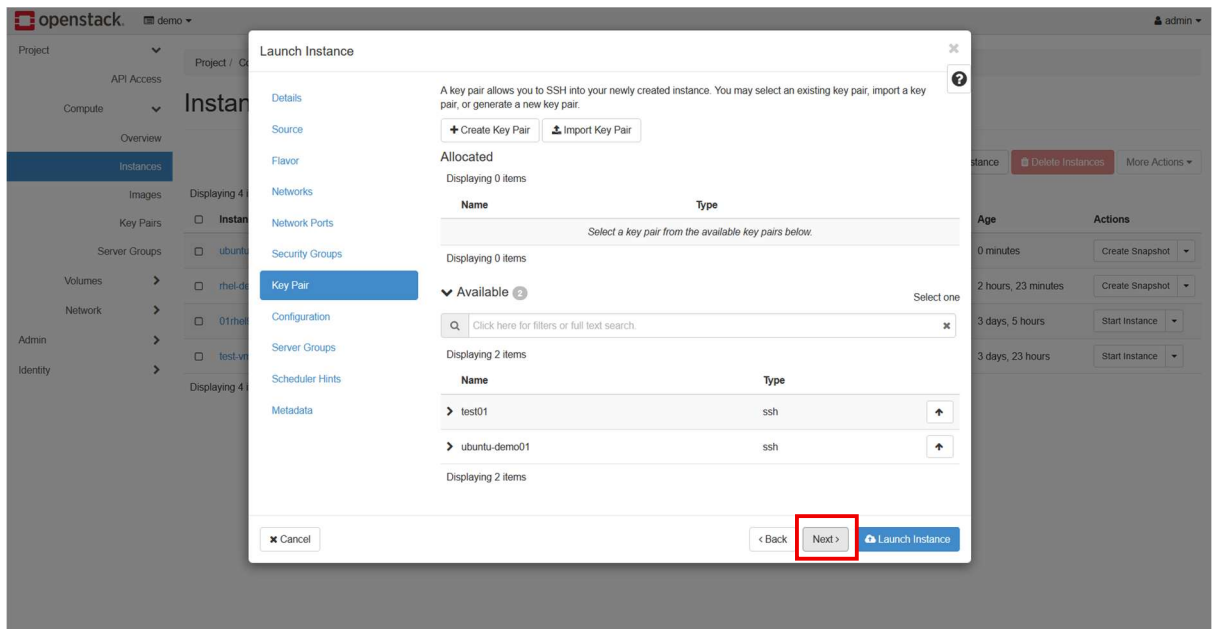
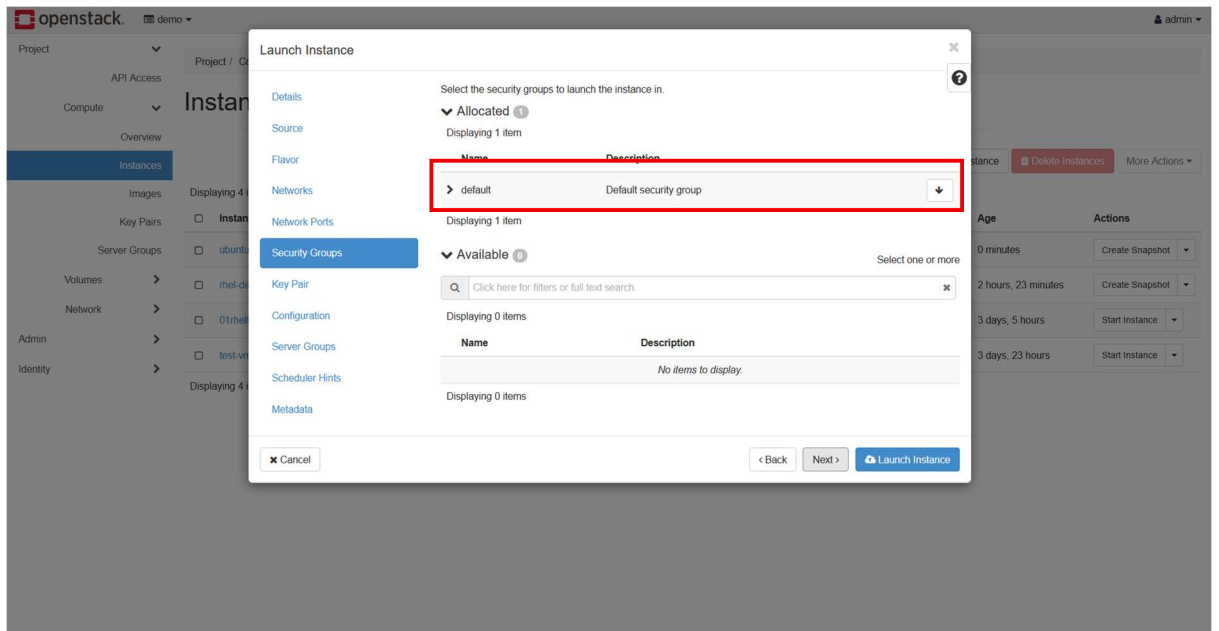
Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
ubuntu-demo01	ubuntu-jammy	10.200.124.197	m1.small	ubuntu-demo01	Active	nova	None	Running	32 minutes	Create Snapshot
rhel-demo02	-	10.200.124.193	ds2G	test01	Active	nova	None	Running	1 hour, 57 minutes	Create Snapshot
01rhel95	-	10.200.124.200	m1.medium	test01	Shutoff	nova	None	Shut Down	3 days, 4 hours	Start Instance
test-vm124-ssh	ubuntu-jammy	10.200.124.191	m1.tiny	test01	Shutoff	nova	None	Shut Down	3 days, 22 hours	Start Instance



The screenshot shows the 'Launch Instance' dialog box in the OpenStack dashboard. The dialog has several tabs: Details, Source, Flavor, Networks, Network Ports, Security Groups, Key Pair, Configuration, Server Groups, Scheduler Hints, and Metadata. The 'Details' tab is active. In this tab, the 'Instance Name' field is highlighted with a red box and contains the text 'ubuntu-demo02'. The 'Count' field is also highlighted with a red box and contains the value '1'. Other fields include 'Project Name' (demo), 'Description', and 'Availability Zone' (nova). A progress indicator on the right shows 'Total Instances (10 Max)' with a 50% completion bar, indicating 4 current usage, 1 added, and 5 remaining. At the bottom of the dialog, the 'Next >' button is highlighted with a red box.







cloud-config.yaml

File Edit View

```
#cloud-config
chpasswd:
  list: |
    ubuntu:password
  expire: False
ssh_pwauth: True
|
```

```
#cloud-config

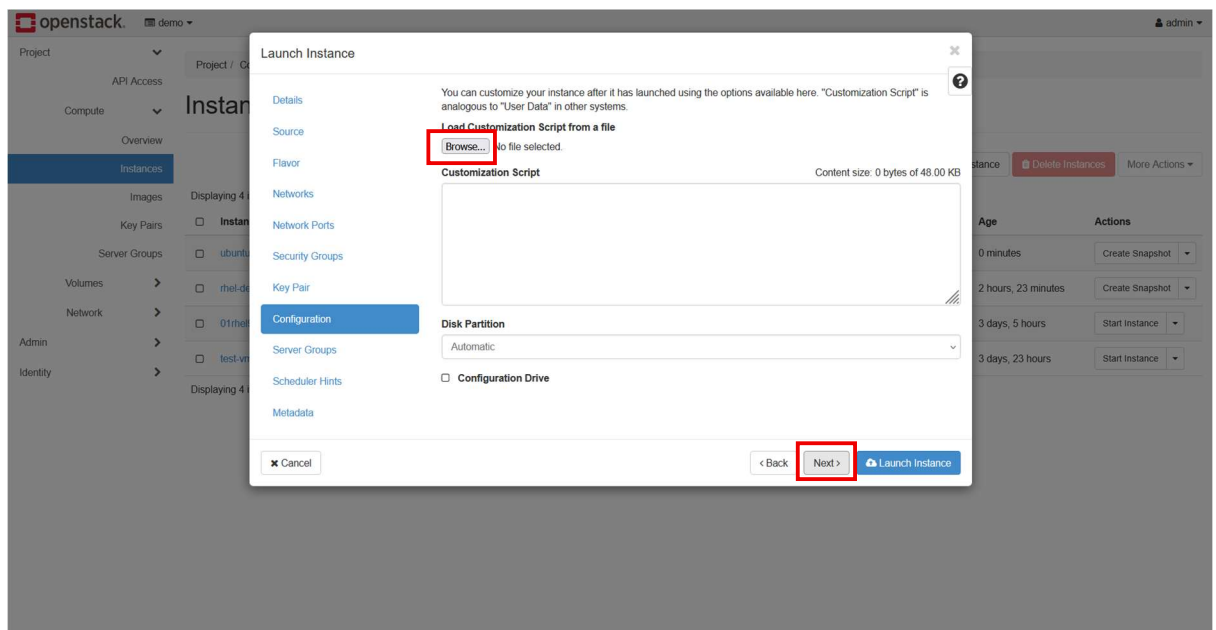
chpasswd:

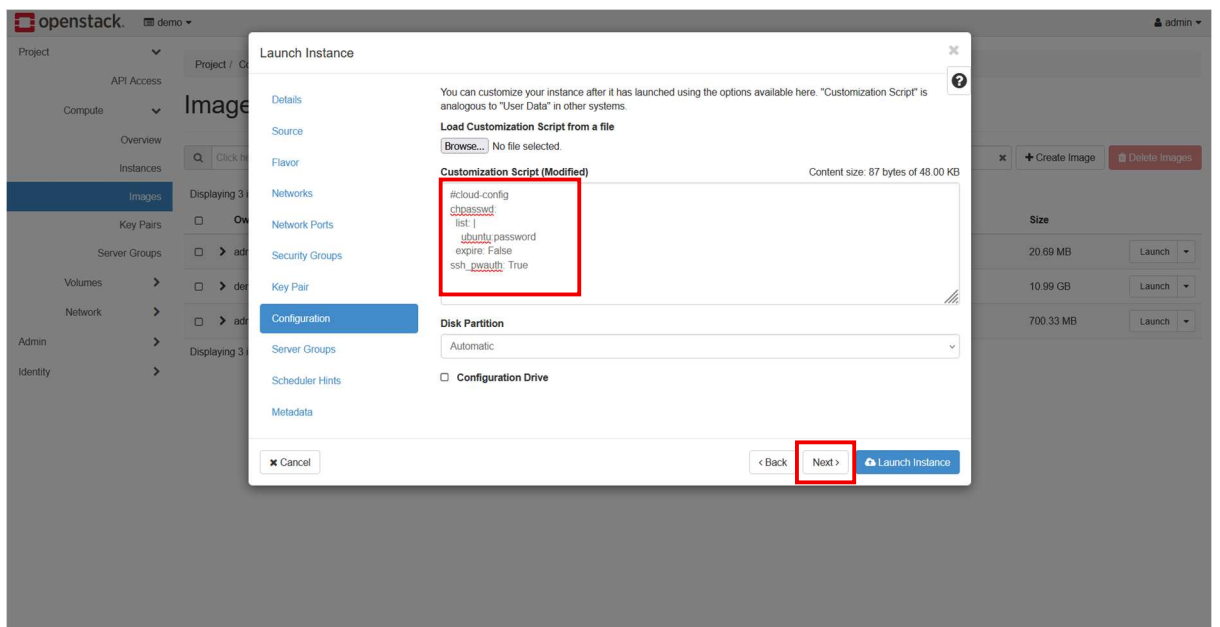
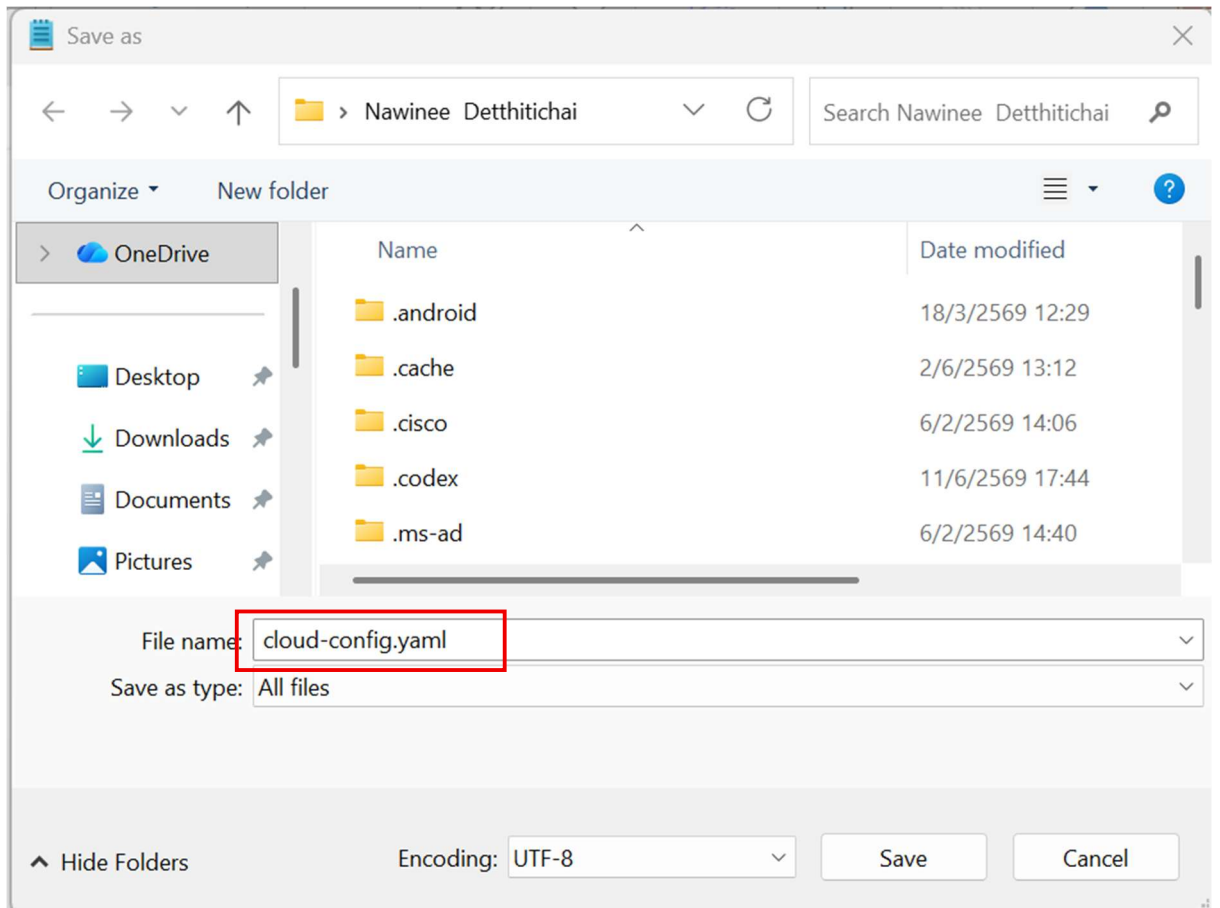
list: |

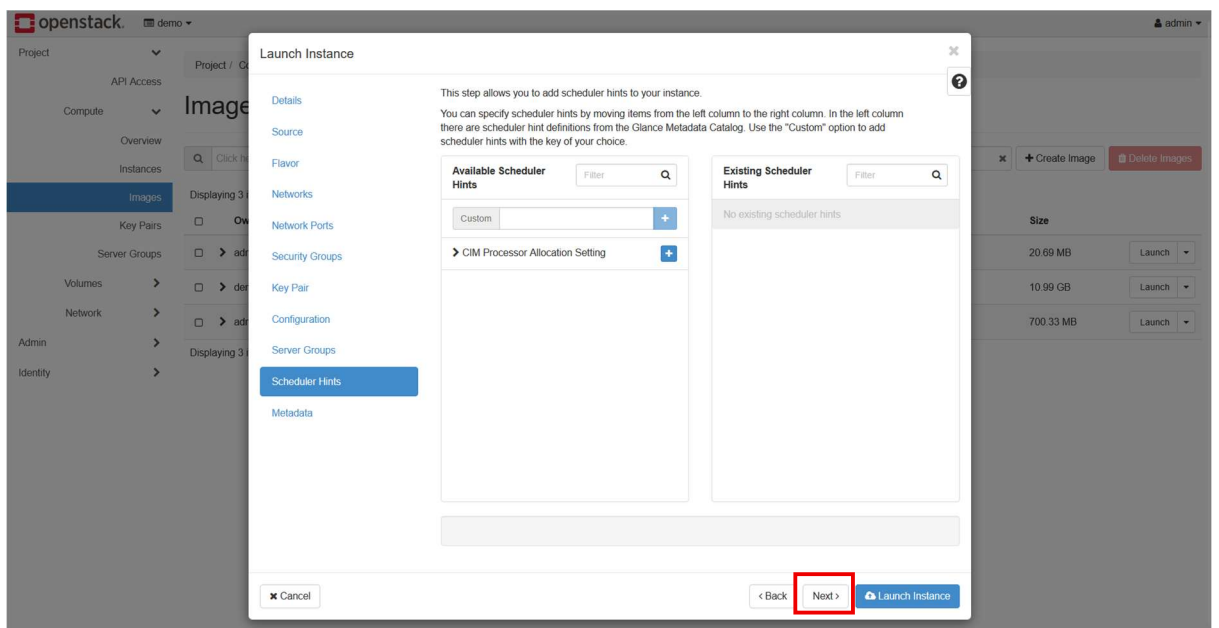
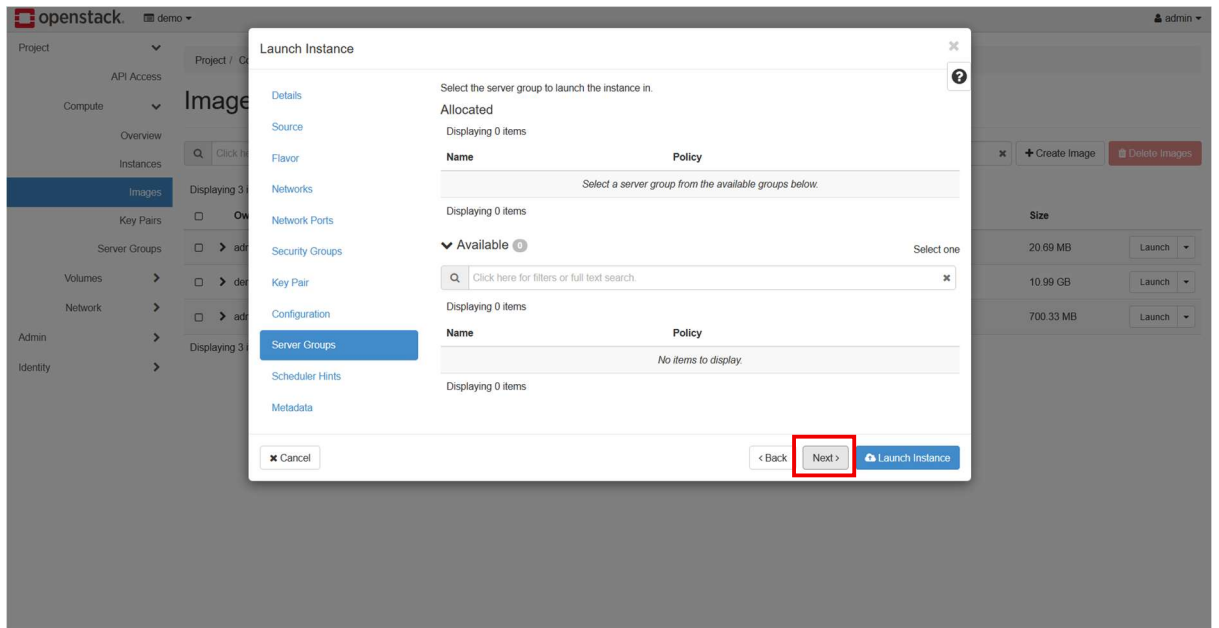
    ubuntu:password

expire: False

ssh_pwauth: True
```







Project

API Access

Compute

Overview

Instances

Images

Key Pairs

Server Groups

Volumes

Network

Admin

Identity

demo

admin

Displaying 5 Items

Instance Name	Image Name	IP Address	Flavor	Key Pair	Status	Availability Zone	Task	Power State	Age	Actions
ubuntu-demo02	-	10.200.124.196	m1.small	-	Active	nova	None	Running	0 minutes	Create Snapshot
ubuntu-demo01	ubuntu-jammy	10.200.124.202	m1.small	ubuntu-demo01	Active	nova	None	Running	13 minutes	Associate Floating IP Attach Interface Detach Interface Edit Instance Attach Volume Detach Volume Update Metadata Edit Security Groups Edit Port Security Groups
rhel-demo02	-	10.200.124.193	ds2G	test01	Active	nova	None	Running	2 hours, 36 minutes	Console
01rhel95	-	10.200.124.200	m1.medium	test01	Shutoff	nova	None	Shut Down	3 days, 5 hours	View Log Rescue Instance Pause Instance Suspend Instance Shelve Instance Resize Instance Lock Instance Soft Reboot Instance Hard Reboot Instance Migrate Instance Shut Off Instance Rebuild Instance Delete Instance
test-vm124-ssh	ubuntu-jammy	10.200.124.191	m1.tiny	test01	Shutoff	nova	None	Shut Down	3 days, 23 hours	

Displaying 5 Items

http://10.200.124.188/dashboard/project/instances/2ff59320-a233-4d3b-a249-847f502586dc/?tab=instance_details_console

Connected to QEMU (instance-00000013)Send Ctrl+Alt+Del

```
Ubuntu 22.04.5 LTS ubuntu-demo02 tty1
ubuntu-demo02 login: ubuntu
Password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-187-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Aug 28 07:11:48 UTC 2026

System load:  0.18           Processes:      88
Usage of /:   8.3% of 19.20GB Users logged in:  0
Memory usage: 9%            IPv4 address for ens3: 10.200.124.196
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ubuntu-demo02:~$ _
```